Plants that evolved under high phylogenetic diversity have higher establishment success, particularly in undisturbed communities: Supporting Information

Joshua I. Brian, Mark van Kleunen, Wayne Dawson, Anne Kempel, Weihan Zhao, Jane A. Catford

Supplementary Methods	2
Phylogenetic tree construction	2
Climate data	3
Robustness tests	3
Species origin	5
Structural equation models for trait inclusion	5
Supplementary methods figures (Figs. S1 – S8)	7
Supplementary methods tables (Table S1)	15
Supplementary Results	19
Effect of absolute indigenous-PD (summary and Figs. S9 – S12)	19
Effect of relative indigenous-PD (summary and Figs. S13 – S16)	24
Effect of precipitation dissimilarity (Fig. S17)	30
Species origin and trait inclusion (summary)	31
Supplementary tables for effect of absolute indigenous-PD (Tables S2 – S5)	32
Supplementary tables for effect of relative indigenous-PD (Tables S6 – S9)	40
Supplementary tables for species origin and traits (Tables S10 – S12)	50
Supplementary References	54

Supplementary Methods

Phylogenetic tree construction

We constructed a phylogenetic tree for the global extant flora following the process used in van Kleunen et al. (2020) and Yang et al. (2021). The basis for this tree was the mega phylogeny for seed plant species built by Smith and Brown (Smith & Brown 2018). We first harmonized the tip names of the mega phylogeny according to the World Checklist of Vascular Plants (WCVP) v.11 and pruned the tree to only include species with accepted names in the WCVP v.11. Tip names matched to synonyms were consolidated with corresponding accepted names. Because 26.7% of the accepted species names in WCVP v.11 (n = 105,749) were missing from the phylogenetic tree, we added them manually using the R package "phytools" (Revell 2024). Specifically, for infraspecific names with their conspecific names in the tree, we added them to the root of their species (n = 42,255, 10.65%). The remaining ones were added to the root of their genus (n = 62,020, 15.63%) or, if the whole genus was missing, to the family root (n = 1,474, 0.37%). We used this phylogenetic tree for the global species pool (all 396,709 seed plant species with accepted names in the WCVP v.11) for the main analysis. A subset of this phylogeny, containing all the sown and resident species in the experiments, is available in the accompanying Zenodo repository ('phylogenetic_hypothesis.RData').

This phylogeny contains both woody and non-woody plants. The presence of woody plants (i.e. trees and large shrubs) could potentially bias our interpretation of indigenous-PD by including species that our focal grassland species do not interact with. We do not believe that this is concerning, as the aim of indigenous-PD is to capture the evolutionary conditions of our focal species. In evolutionary time, there is strong potential for woody plants to have affected grassland plants, even if there was not direct species-species interaction. For example, higher woody species diversification narrows the range of resources available in a given area for non-woody species, encouraging specialisation and competitive superiority in remaining niche space among those non-woody species (Stachowicz & Tilman 2005). Further, evidence from the geological record (e.g. in both Europe and Madagascar) suggests tree and grassland cover oscillated through time, implying interactions and shifts in competitive dominance between functional groups at habitat edges (Bond & Midgley 2012; Silander Jr. et al. 2024; Czyżewski et al. 2026). The evolution of grasses potentially occurred

in mosaic landscapes such as in forest gaps, and C3 grasses continue to be more commonly found in forest understory than in open grassland (Scholes & Archer 1997; Strömberg 2011). Finally, there is contemporary evidence from competition experiments that trees and grassland species continue to compete (e.g. Belsky 1994; Ludwig et al. 2004; Riginos 2009; Dohn et al. 2013). Therefore, we believe woody species are important to maintain in our analyses of indigenous-PD as they likely played a role in the evolution of our grassland species, whether through direct interactions (competition, facilitation) or indirectly through the constriction of available niche space for non-woody species.

Nevertheless, to test whether woody species affect overall indigenous-PD, we calculated what proportion of species in each species' native range were tree species. We did this by using the list of tree species and distributions from the GlobalTreeSearch database from Botanic Gardens Conservation International (BGCI 2026). We tested whether there was a significant relationship between the proportion of tree species in a species' native (i.e. home) range and mean alpha PD for each species. If there was a significant relationship, this would imply that species indigenous-PD is influenced by the diversity of tree species in species' native (home) ranges. However, there was no significant relationship ($F_{1, 161} = 0.629$, $p = 0.429$; Fig. S6). Therefore, calculations of indigenous-PD were not sensitive to tree species richness.

Climate data

As an alternative to the PCA of all bioclimatic variables, we also calculated temperature and precipitation dissimilarity using two bioclimatic variables directly: minimum temperature of the coldest month (bio6) and precipitation of the driest quarter (bio17). These two variables loaded most strongly on to PC1 and PC2, respectively, and are both considered key variables determining colonisation likelihood in central European plant communities (Dullinger et al. 2017). We used these two dissimilarities in alternate models to check robustness of key trends (see *Supplementary analyses: Robustness tests*).

Robustness tests

To test the robustness of our analyses, we tested sets of alternate models. We had three sets of alternate models for testing the influence of absolute indigenous-PD. First, as several trends were consistent across models testing both absolute and relative indigenous-PD, we repeated

analyses only using Kempel and Müller data ('K+M' models), to confirm that significant results in the absolute models were not being driven purely by Haeuser data. Second, we calculated a coarse version of 'difference in PD' (not abundance-weighted) between planted species and the resident community, using Haeuser and Kempel data, and used this metric as the key response variable ('K+H' models). This was possible as Kempel had semi-quantitative resident vegetation data at the site level and was used to check whether the absolute results were consistent for difference in PD rather than raw PD metrics. Third, we reran the main models using the single climate variables ('SC' models) rather than the dissimilarities from the principal components analysis.

We had two sets of alternate models for the testing the influence of relative indigenous-PD. First, we ran SC models as described above. Second, we ran all models using raw indigenous-PD metrics rather than difference in PD ('Hnd' models). Results for all alternate models generally supported the main results; all models are compared in Tables S2 – S9. For all alternate models, we ran them for all four indigenous-PD metrics (mean alpha PD, maximum alpha PD, median alpha PD, gamma PD).

To enable comparison of effect size coefficients, we carried out all analyses twice; first with unstandardised continuous variables, and second with standardised continuous variables (centred on the mean and divided by two standard deviations; Gelman 2008). We report both for transparency in the supplementary tables S2 – S9, with the standardised effect size in brackets. To fairly compare the effect of different variables (i.e. Figs. 2, 4 in main text), we use the standardised variables, but we maintain the unstandardised variables when interpreting the effect of indigenous-PD alone (i.e. Figs. 3, 5, 6 in main text) as the values have specific ecological meanings. Standardising the continuous variables did not alter p-values, except in very rare cases where two standardised variables were in an interaction. The p-value of the interaction did not change, only the p-values of the main effect coefficients for those variables. In these cases, we also report the standardised p-values in brackets in tables S2 – S9. As the main effect terms are not interpreted when the variables are in interactions, overall interpretations are identical between the standardised and non-standardised versions.

Species origin

To test whether our results held for both native and non-native sown species, we added species origin (native/non-native) to all our final selected models as a fixed effect and compared models with and without species origin using difference in AIC. Models within 2 AIC were considered equivalent. See Table S10 for results.

To test whether our results held for species from different biogeographic regions, we assigned species to their native 'biogeographic syndrome' following Fristoe et al. (2023). Briefly, biogeographic syndromes provide information on the combination of continents to which a species is native. For example, a species that is only native somewhere in Europe would have a different biogeographic syndrome to a species that is native to both Europe and Asia, etc. Overall, our 166 species showed 24 different biogeographic syndromes. We added species biogeographic syndrome to all our final selected models as a random effect and compared models with and without biogeographic origin using difference in AIC. Models within 2 AIC were considered equivalent. See Table S10 for results.

We also tested whether the PD-trait correlations were affected by biogeographic syndrome. We carried out two additional tests: first, we limited data to those species that were native to North America or Europe, to avoid potential biases introduced by tropical species that can have systematically different trait values (Moles et al. 2007) but are also systematically different in regional indigenous-PD (Fig. 1c, main text). Second, we used the full dataset but included biogeographic syndrome as a random variable. See Table S12 for results.

Structural equation models for trait inclusion

To investigate whether the effect of indigenous-PD was mediated entirely through plant traits, we ran two sets of structural equation models for the main models where indigenous-PD was found to be significant. In the first set, we limited indigenous-PD to have only indirect effects on the response variable (i.e. the effect of indigenous-PD is mediated entirely through plant traits), while in the second set we allowed both direct and indirect effects of indigenous-PD (Fig. S8). Models were identical to the main text models aside from the addition of traits (i.e. all other covariates always remained in the models), as the goal was to verify whether the

main text model results were robust to the inclusion of traits. The model sets were compared using AIC, with a lower AIC indicating a superior model.

We chose three traits for inclusion: specific leaf area (SLA), vegetative height, and seed mass. These three traits all load strongly on to the aboveground spectrum of plant form and function (Díaz et al. 2016; Carmona et al. 2021) and are the three traits most measured for naturalised plant species (Grenié et al. 2025). They were also the three traits for which we had the most data for our 166 species (Table S11). After exploring all data sources, we had trait data for 153 of our 166 species (seed mass), 149 species (vegetative height) and 133 species (SLA). We chose to limit our analyses to species that had trait data, rather than imputing missing values, for two reasons. First, while imputation is useful for community-level trends and multivariate analyses, it is less appropriate when specific data points that were created by imputation would be used in downstream analysis, as imputation methods are not accurate enough to produce reliable species-level data (Gorné et al. 2025). Second, imputation requires the incorporation of phylogenetic information. However, indigenous-PD has a phylogenetic signal - in other words, more closely related species are more likely to have more similar indigenous-PD values ($C_{mean} = 0.157$, $p = 0.002$; Moran's $I = 0.063$, $p = 0.006$). Therefore, imputation will generate circularity: if we use phylogenetic information to help impute missing trait values, and then ask if the effect of indigenous-PD on success is mediated through traits, we are more likely to find a positive result, because we are contributing to a phylogenetic signal in the traits that is independently seen in indigenous-PD.

For each set of structural equation models, we tested two trait combinations. First, we tested the effect of SLA and height together (N = 131 species for both traits), as representatives of the two axes of aboveground form and function (Carmona et al. 2021). Second, we tested the effect of SLA, height and seed mass together (N = 121 species for all three traits), to incorporate reproductive investment.

Models were fit using the piecewiseSEM R package (Lefcheck 2016). See Table S11 for results.

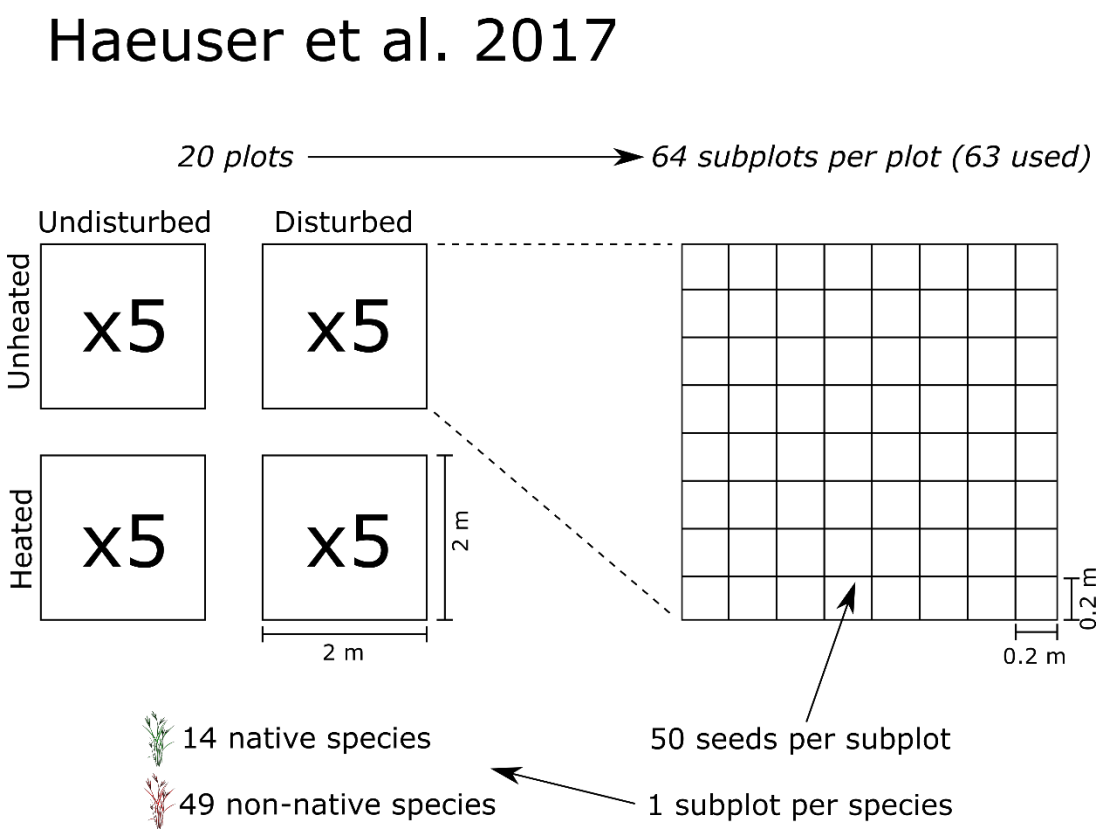


Figure S1: Experimental design of Haeuser et al. 2017. Replicates of each treatment combination were arranged within a single site at the University of Konstanz, Germany.

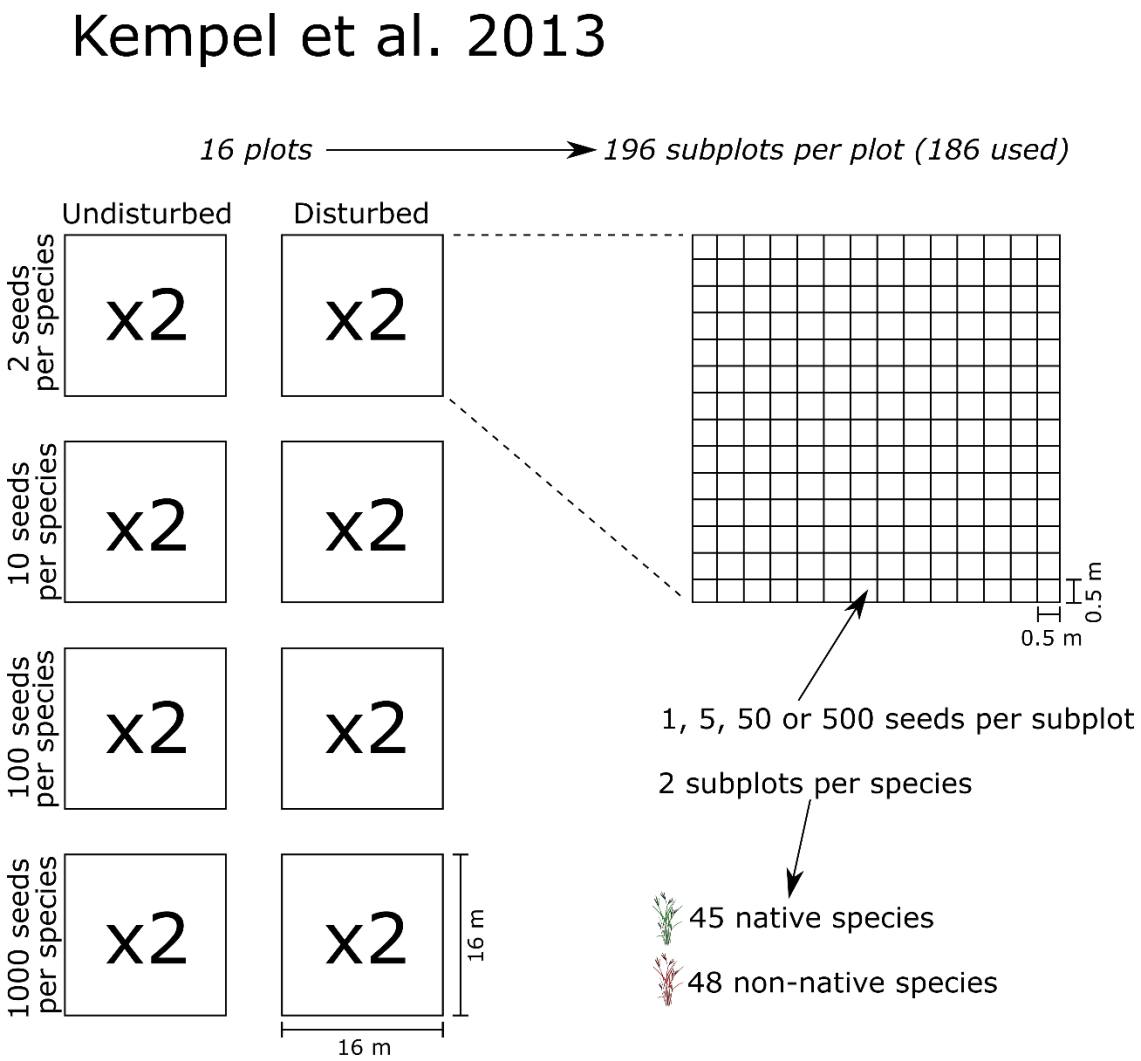


Figure S2: Experimental design of Kempel et al. 2013. Replicates of each treatment combination were arranged across 16 different sites in the northern half of the canton of Bern (Switzerland). Sites were a maximum of 50 km from each other.

Müller et al. 2016

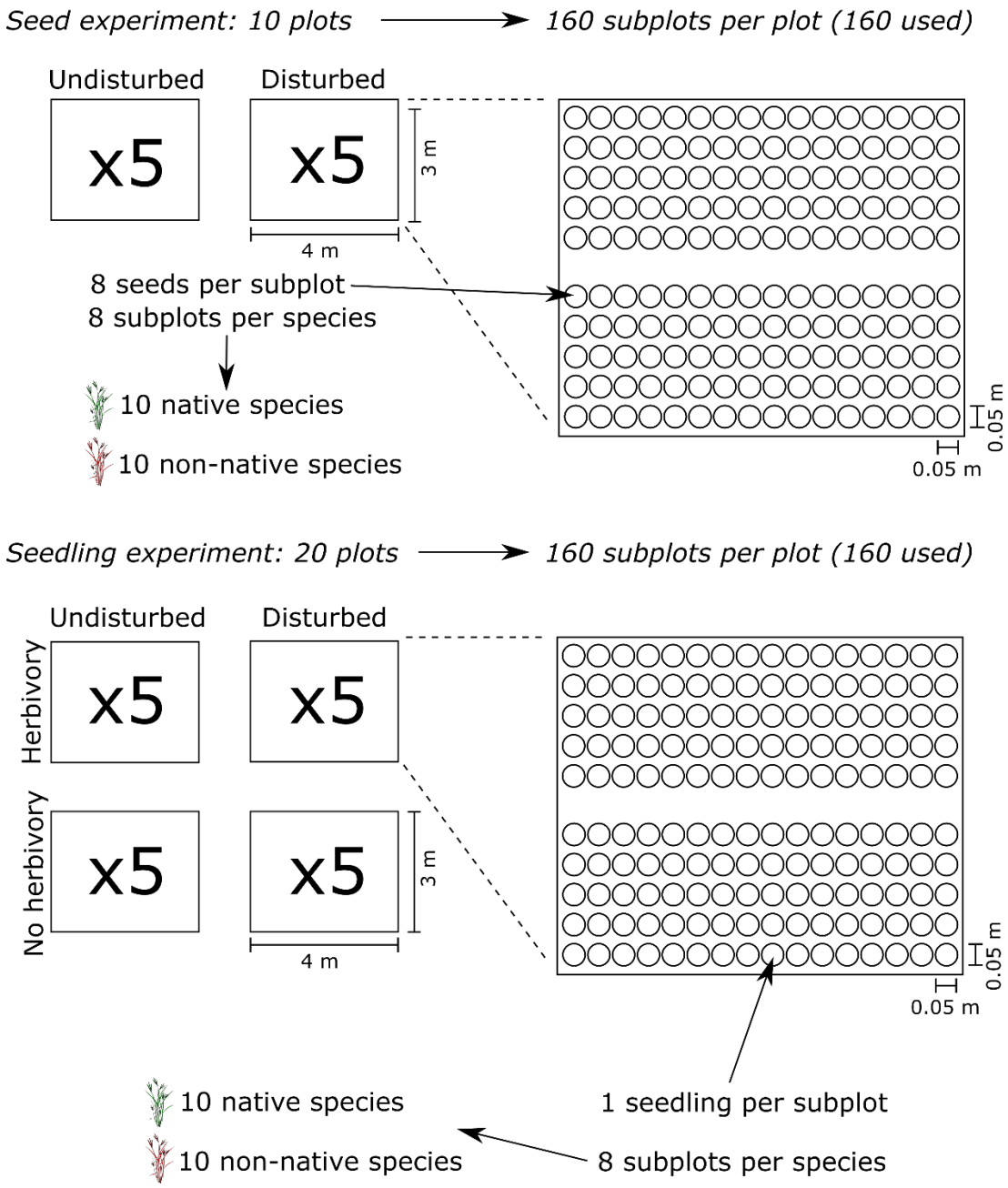


Figure S3: Experimental design of Müller et al. 2016. Replicates of each treatment combination were arranged across five sites adjacent to the University of Konstanz, Germany. Note that both Müller experiments also featured a belowground biocide treatment; only control plots with respect to this variable are used in the present study. The experimental species were identical between the seed and seedling experiments, with one exception in the native species. Therefore, while each experiment used 20 species, in total 21 species were used (11 native, 10 non-native).

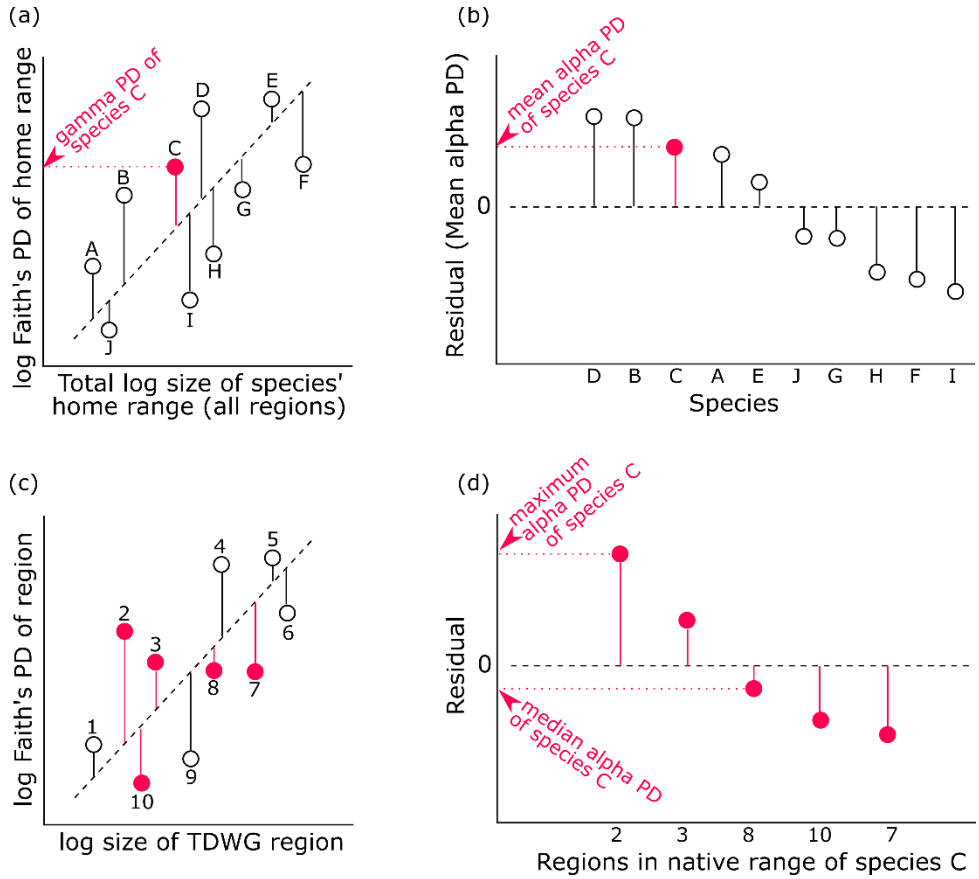


Figure S4: Calculating metrics of phylogenetic diversity (PD). (a) Gamma PD is defined as the total PD in the indigenous range of a species, across all TDWG3 regions. Here, ten species are plotted (species A – species J), with the total phylogenetic diversity in the native range of species C highlighted (the gamma PD of species C). (b) Mean alpha PD (the focal metric in the study) is defined as the mean PD across the indigenous range of a species. It is calculated as the residual of the log-log relationship between native range size and gamma PD (species C highlighted as an illustration). (c) Individual TDWG3 regions are plotted and regressed against the PD within each region, with ten regions shown as an example. Species C is native to five of those regions (highlighted). (d) Maximum alpha PD is defined as the maximum area-corrected PD of a region in which a species is indigenous (the highest residual of all the regions where it is native), while median alpha PD is defined as the median area-corrected PD of a region in which a species is indigenous. Calculation of mean alpha, maximum alpha and gamma PD were as described in Fristoe et al. (2023).

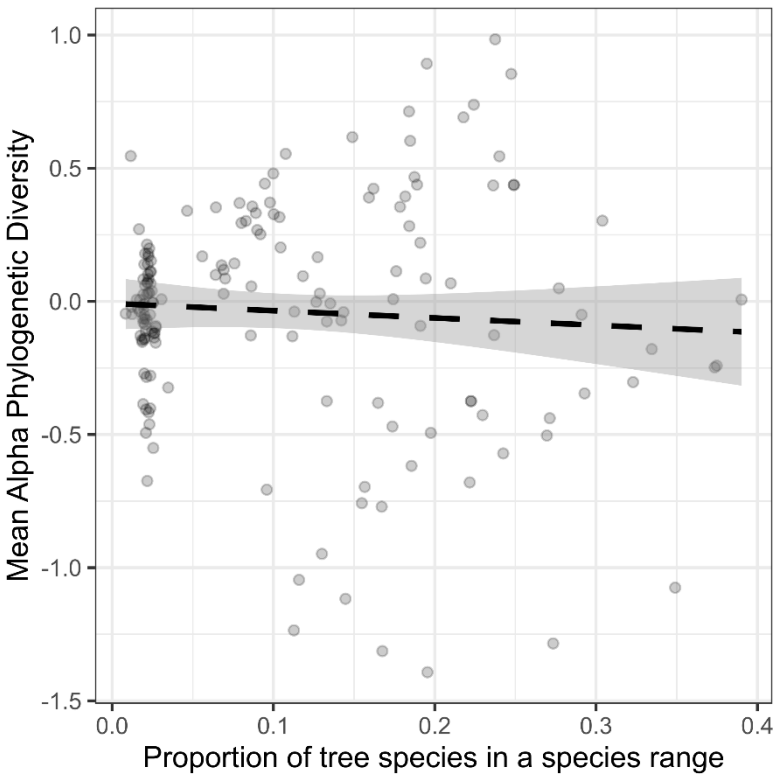


Figure S5: The indigenous-PD for a given species is not related to the proportion of tree species in its indigenous range (p = 0.429). Each dot is a species. Note that the 95% confidence interval (grey shaded area) overlaps zero for the whole range of the x-axis suggesting that, for any given tree proportion, all species with that proportion in their indigenous range still have a collective mean indigenous-PD of 0, as predicted by the null hypothesis. Collectively, this strongly suggests that tree species do not bias calculations of indigenous-PD. The large collection of points at approximately 0.025 on the x-axis is species of European origin.

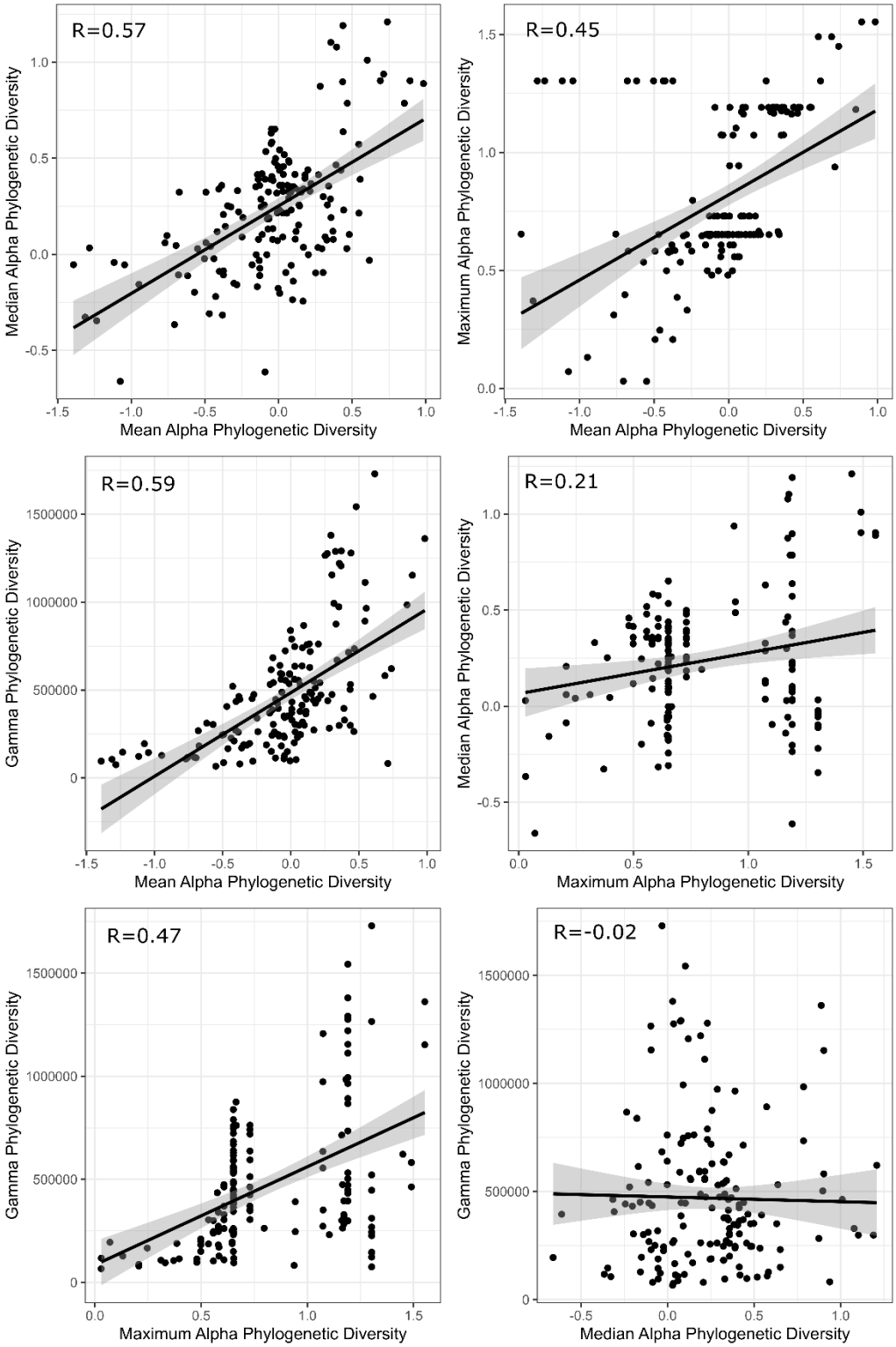


Figure S6: Correlations between the phylogenetic diversity metrics used in the study. Inset values indicate Pearson's R. Each point is a species.

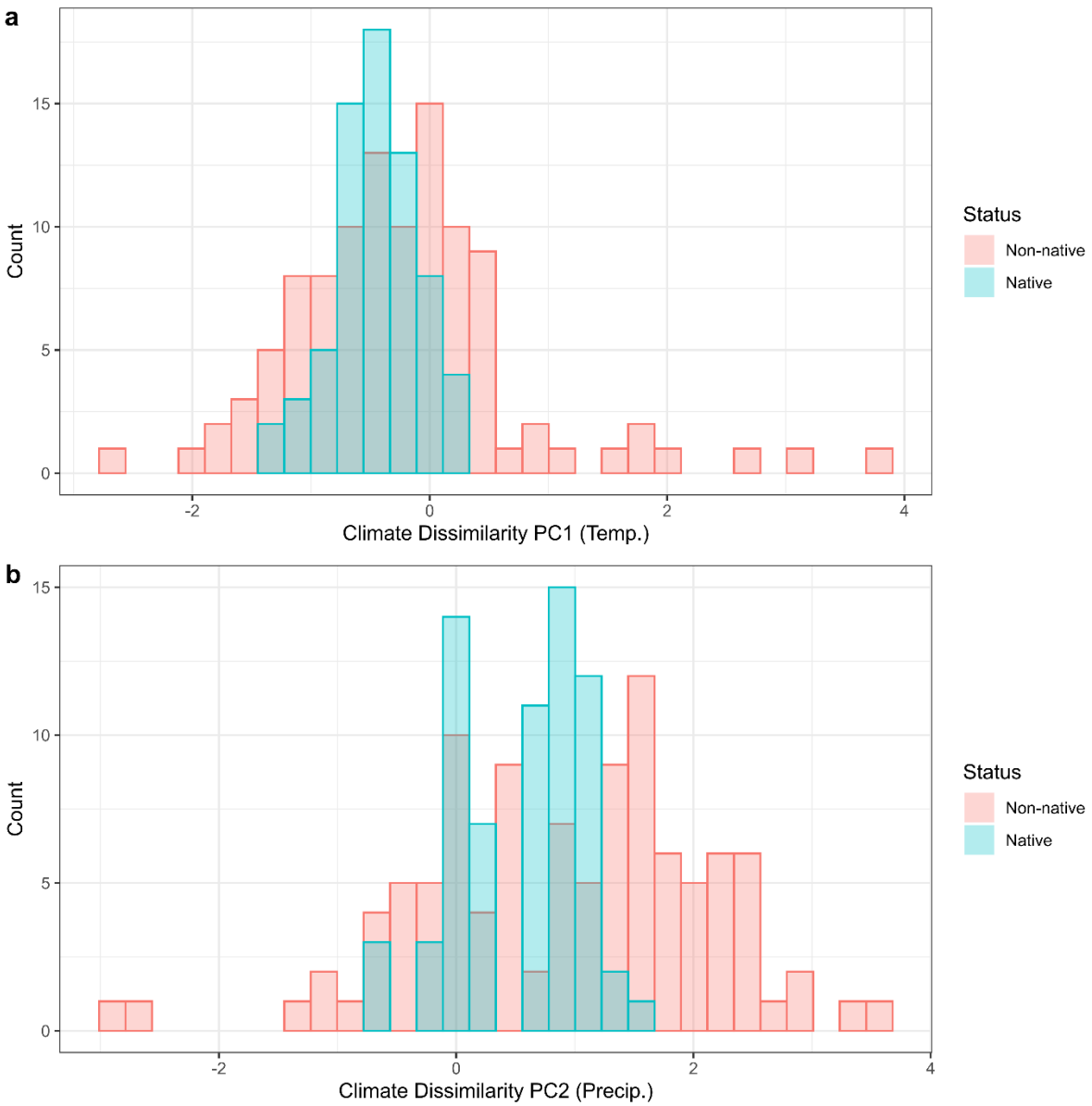


Figure S7: Histograms showing the climate dissimilarity (experimental climate minus native range climate) for all species in the experiments, where 'count' refers to the number of species in each climate bin. Species are divided by native status (native or non-native). Non-native sown species had higher dissimilarities than native sown species (temperature dissimilarity, $\chi^2_1 = 7.296$, $p=0.007$; precipitation dissimilarity, $\chi^2_1 = 21.945$, $p<0.001$), and native species had dissimilarities more closely centred on zero.

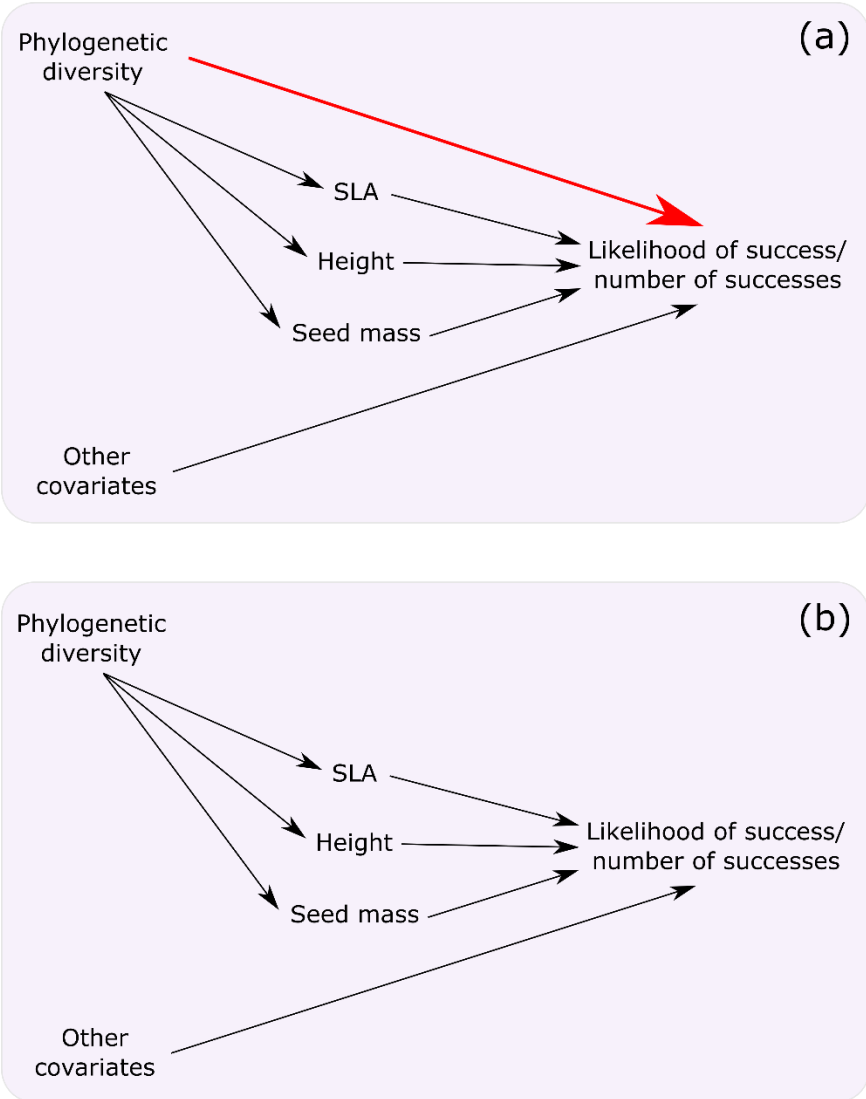


Figure S8: Structure of the two competing structural equation models (SEMs) tested for each model where indigenous phylogenetic diversity (indigenous-PD) was found to be significant. (a) SEM testing both direct and indirect effects of indigenous-PD on the likelihood of planted species success and the number of successes. (b) SEM testing indirect effects only of indigenous-PD on species success, whereby the effect of indigenous-PD is mediated entirely through species traits. Models (a) and (b) were compared with AIC with the lower AIC being preferred. The link being tested by this approach is highlighted in red; if model (a) has a lower AIC this indicates that the PD has a direct effect on species success in the experiment, independent of the tested traits. 'Other covariates' includes all other variables included in the main text models to ensure consistency in interpretation.

Supplementary methods tables

Table S1: Complete list of all species across all three experiments. 'Species' uses the Plants of the World (POWO) accepted name. 'ID' is the identification number in the World Checklist of Vascular Plants (WCVP) database. The three study columns indicate whether a species was sown (1) or not sown (0) in that study. 'Status' indicates whether the species was native or non-native to the sown location.

Species	**ID**	**Müller**	**Haeuser**	**Kempel**	**Status**
Achillea filipendulina	3109923	0	1	1	Exotic
Achillea millefolium	2908230	1	1	0	Native
Achillea nobilis	3110044	1	0	0	Native
Aconitum carmichaelii	2618501	0	1	0	Exotic
Ajuga reptans	5385	0	1	1	Native
Alcea rosea	2627710	0	0	1	Exotic
Allium schoenoprasum	296525	0	1	0	Native
Alyssum alyssoides	2631899	0	0	1	Native
Amaranthus tricolor	2633132	0	1	0	Exotic
Anchusa arvensis	2636464	0	0	1	Native
Anchusa capensis	2636491	0	0	1	Exotic
Anchusa officinalis	2636606	0	0	1	Native
Anoda cristata	2641086	0	0	1	Exotic
Antirrhinum majus	2642724	0	1	0	Exotic
Aquilegia viridiflora	2645036	0	0	1	Exotic
Aquilegia vulgaris	2645045	0	1	1	Native
Arabis hirsuta	2645681	0	0	1	Native
Arabis caucasica	2645447	0	0	1	Exotic
Aster amellus	3099441	1	0	0	Native
Aster bellidiastrum	3099482	0	0	1	Native
Atropa bella-donna	2665943	0	1	0	Native
Aurinia saxatilis	2666583	0	0	1	Exotic
Borago officinalis	2679695	0	0	1	Native
Bunias orientalis	2686094	0	0	1	Exotic
Calendula officinalis	2910167	0	0	1	Exotic
Calystegia sepium	483559	0	0	1	Native
Campanula armena	363990	0	0	1	Exotic
Campanula barbata	364044	0	0	1	Native
Campanula pyramidalis	365752	0	0	1	Exotic
Campanula rapunculus	365813	0	0	1	Native
Campanula rotundifolia	365918	0	0	1	Native
Cardamine pratensis	2700274	1	0	1	Native
Cenchrus caudatus	487074	0	1	0	Exotic
Centaurea macrocephala	3050068	0	1	0	Exotic
Cerastium tomentosum	2710954	1	1	0	Exotic
Chelidonium majus	2716702	0	0	1	Native
Cichorium intybus	2901597	0	0	1	Native
Circaea lutetiana	2721837	0	0	1	Native

Clarkia amoena	2724644	0	0	1	Exotic
Clematis terniflora	2726804	0	0	1	Exotic
Clematis vitalba	2726964	0	0	1	Native
Convolvulus arvensis	483761	0	0	1	Native
Convolvulus tricolor	484499	0	0	1	Exotic
Cuminum cyminum	2747364	0	1	0	Exotic
Cynoglossum amabile	2751851	0	0	1	Exotic
Cynoglossum officinale	2752089	0	0	1	Native
Datura stramonium	2757848	0	0	1	Exotic
Delphinium ajacis	2759384	0	1	0	Exotic
Dianthus armeria	2763820	0	0	1	Native
Dianthus caryophyllus	2764005	0	0	1	Exotic
Digitalis lanata subsp. trojana	3008772	0	1	0	Exotic
Diplotaxis tenuifolia	2771347	1	0	0	Exotic
Echium vulgare	2784423	0	0	1	Native
Epilobium dodonaei	2790362	1	0	1	Native
Epilobium tetragonum	2791119	1	0	0	Native
Eragrostis trichodes	413314	0	1	0	Exotic
Eranthis hyemalis	2791911	0	1	0	Exotic
Erigeron acris	3104709	0	0	1	Native
Eritrichium canum	2797689	0	1	0	Exotic
Eschscholzia californica	2801972	0	0	1	Exotic
Eudianthe coeli-rosa	2802809	0	0	1	Exotic
Eupatorium cannabinum	2926599	1	0	0	Native
Euthamia graminifolia	3102167	1	0	0	Exotic
Galeopsis angustifolia	85331	0	0	1	Native
Gilia tricolor	2825947	0	1	0	Exotic
Gypsophila elegans	2836649	0	0	1	Exotic
Gypsophila paniculata	2836785	1	0	0	Exotic
Helenium bigelovii	3115764	0	1	0	Exotic
Helianthus annuus	2910966	0	0	1	Exotic
Helianthus debilis	3120900	0	1	0	Exotic
Heliotropium arborescens	2843787	0	1	0	Exotic
Hesperis matronalis	2847567	0	1	0	Exotic
Hibiscus trionum	2850654	0	0	1	Exotic
Hordeum jubatum	419358	0	1	0	Exotic
Iberis amara	2859697	0	0	1	Native
Iberis sempervirens	2859919	0	0	1	Exotic
Ipomoea tricolor	482807	0	0	1	Exotic
Iris domestica	334601	0	1	0	Exotic
Iris sibirica	322377	0	1	0	Native
Ismelia carinata	3112289	0	0	1	Exotic
Jacobaea maritima subsp. bicolor	3082659	0	0	1	Exotic
Jacobaea vulgaris	3082777	1	0	0	Native
Knautia arvensis	2334543	0	0	1	Native
Lathyrus odoratus	2351328	0	0	1	Exotic

Legousia speculum-veneris	358587	0	0	1	Native
Lepidium graminifolium	2337784	1	0	0	Native
Lepidium heterophyllum	2337794	1	0	0	Exotic
Leucanthemum vulgare	3112766	0	0	1	Native
Lilium formosanum	293214	0	1	0	Exotic
Lilium regale	279843	0	1	0	Exotic
Lithotoma axillaris	494315	0	1	0	Exotic
Lobelia erinus	353498	0	0	1	Exotic
Lobelia inflata	353345	0	1	0	Exotic
Lobularia maritima	2338457	0	0	1	Exotic
Lotus corniculatus	2345329	0	1	0	Exotic
Lupinus ehrenbergii var. ehrenbergii	3017330	0	0	1	Exotic
Lysimachia clethroides	2493128	0	1	0	Exotic
Lysimachia punctata	2492220	0	1	0	Exotic
Lysimachia vulgaris	2492070	0	1	0	Native
Malva alcea	2503573	0	0	1	Native
Malva moschata	2503483	0	0	1	Native
Malva neglecta	2503504	0	0	1	Native
Malva trimestris	2504047	0	0	1	Exotic
Medicago lupulina	2368448	0	0	1	Native
Mentha spicata	125402	0	1	0	Exotic
Monarda fistulosa	129185	0	1	0	Exotic
Monarda punctata	129232	0	1	0	Exotic
Nemophila maculata	2379420	0	1	0	Exotic
Nepeta racemosa	134920	0	1	0	Exotic
Nicotiana mutabilis	2382251	0	1	0	Exotic
Nicotiana sylvestris	2385612	0	1	1	Exotic
Nigella arvensis	2381639	0	0	1	Native
Oenothera biennis	2399189	1	0	0	Exotic
Oenothera glazioviana	2398876	1	0	1	Exotic
Oenothera macrocarpa	2398993	0	0	1	Exotic
Origanum vulgare	143954	0	1	0	Native
Papaver betonicifolium	2502572	0	0	1	Exotic
Papaver cambricum	2560601	0	0	1	Exotic
Papaver dubium	2560793	0	0	1	Native
Papaver orientale	2561370	0	0	1	Exotic
Papaver rhoeas	2561554	0	0	1	Native
Papaver commutatum	2560637	0	0	1	Exotic
Persicaria capitata	2576526	0	1	0	Exotic
Persicaria virginiana	2572324	0	1	0	Exotic
Petunia integrifolia	2549707	0	1	0	Exotic
Phaseolus coccineus	2537776	0	0	1	Exotic
Phleum pratense	433846	0	1	0	Native
Phlox drummondii	2567292	0	0	1	Exotic
Physalis peruviana	2549655	0	0	1	Exotic
Phyteuma orbiculare	355252	0	0	1	Native

Plantago media	2569797	0	1	0	Native
Platycodon grandiflorus	354641	0	1	1	Exotic
Plectocephalus americanus	3046533	0	1	0	Exotic
Polemonium caeruleum	2566071	0	0	1	Native
Potentilla atrosanguinea	2953596	0	1	0	Exotic
Pseudo-fumaria lutea	2561323	0	1	0	Exotic
Rudbeckia fulgida	2928986	0	1	0	Exotic
Rudbeckia triloba	3123672	0	1	0	Exotic
Rumex crispus	2424294	0	1	0	Native
Salpiglossis sinuata	2579347	0	1	0	Exotic
Salvia argentea	182005	0	0	1	Exotic
Salvia farinacea	182630	0	0	1	Exotic
Salvia glutinosa	182731	0	0	1	Native
Salvia lyrata	183102	0	0	1	Exotic
Salvia verticillata	184133	0	1	0	Exotic
Saponaria officinalis	2484914	0	0	1	Native
Scabiosa columbaria	2593679	0	0	1	Native
Senecio ovatus	3082007	0	0	1	Native
Silene chalcedonica	2485408	0	0	1	Exotic
Silene flos-cuculi	2485564	0	0	1	Native
Silene latifolia	2488689	1	1	0	Native
Silene viscosa	2487795	1	0	0	Native
Solanum dulcamara	2932815	0	0	1	Native
Solanum nigrum	2915955	0	0	1	Native
Solidago canadensis	2911023	1	1	0	Exotic
Solidago ptarmicoides	3101706	0	1	0	Exotic
Symphyotrichum lanceolatum	3103441	1	0	0	Exotic
Symphyotrichum novi-belgii	3103597	1	0	0	Exotic
Thymus × citriodorus	204618	0	0	1	Exotic
Thymus pulegioides	205310	0	0	1	Native
Verbena rigida	212652	0	1	0	Exotic
Vicia sepium	2451888	0	1	0	Native
Viola odorata	2459312	0	1	0	Exotic
Viola tricolor	2461201	0	1	0	Native
Zinnia angustifolia	3124687	0	0	1	Exotic
Zinnia peruviana	3124740	0	1	0	Exotic

Supplementary Results

In all cases, models are discussed with reference to the models presented in the main text (i.e. do the additional models support or refute the models using mean alpha PD).

Effect of absolute indigenous-PD

The PD*disturbance interaction was consistent across colonisation and first year survival when using both maximum and median alpha PD (Figs. S9a-c; S10a-c); and was also seen in the second growing season for gamma PD (Fig. S11b). The PD*disturbance interaction was maintained regardless of whether the 'K+M', 'K+H' or 'SC' models were considered (Tables S2-S4), with occasional exceptions in the gamma PD models (Table S5). The PD*disturbance interaction in the second growing season was also seen in the 'SC' models for mean, maximum and gamma PD (Fig. S12).

The positive interaction between mean alpha PD and temperature dissimilarity in predicting second year survival was also seen when using median alpha PD and gamma PD (Figs. S10f, g; S11c). Interestingly, the trend was reversed for first year survival (Figs. S10e; S11a), suggesting that species initially require a level of ecological similarity to the resident community to survive and benefit from higher indigenous-PD in the first growing season, but for species that reach this 'minimum criteria', they are more likely to then succeed and benefit from high PD in the second growing season if they are ecologically different. However, note these effects were generally not seen in the 'SC' models (Tables S2, S4, S5). There were also occasional interactions with precipitation dissimilarity for both maximum alpha PD (Fig. S9d) and median alpha PD (Fig. S10d) in the second growing season. Having a higher indigenous-PD increased the *likelihood of any plants surviving* if precipitation dissimilarity was *low* (Fig. S9d), but having a higher indigenous-PD increased the *number of plants surviving* if precipitation dissimilarity was *high* (Fig. S10d). Though these interactions were only occasionally observed, this provides further evidence for the same interpretation: for a species to be able to survive and benefit from high indigenous-PD (increased likelihood) it needs to be ecologically similar, but for those species that can survive, they derive the greatest benefit (increased numbers) from being ecologically different.

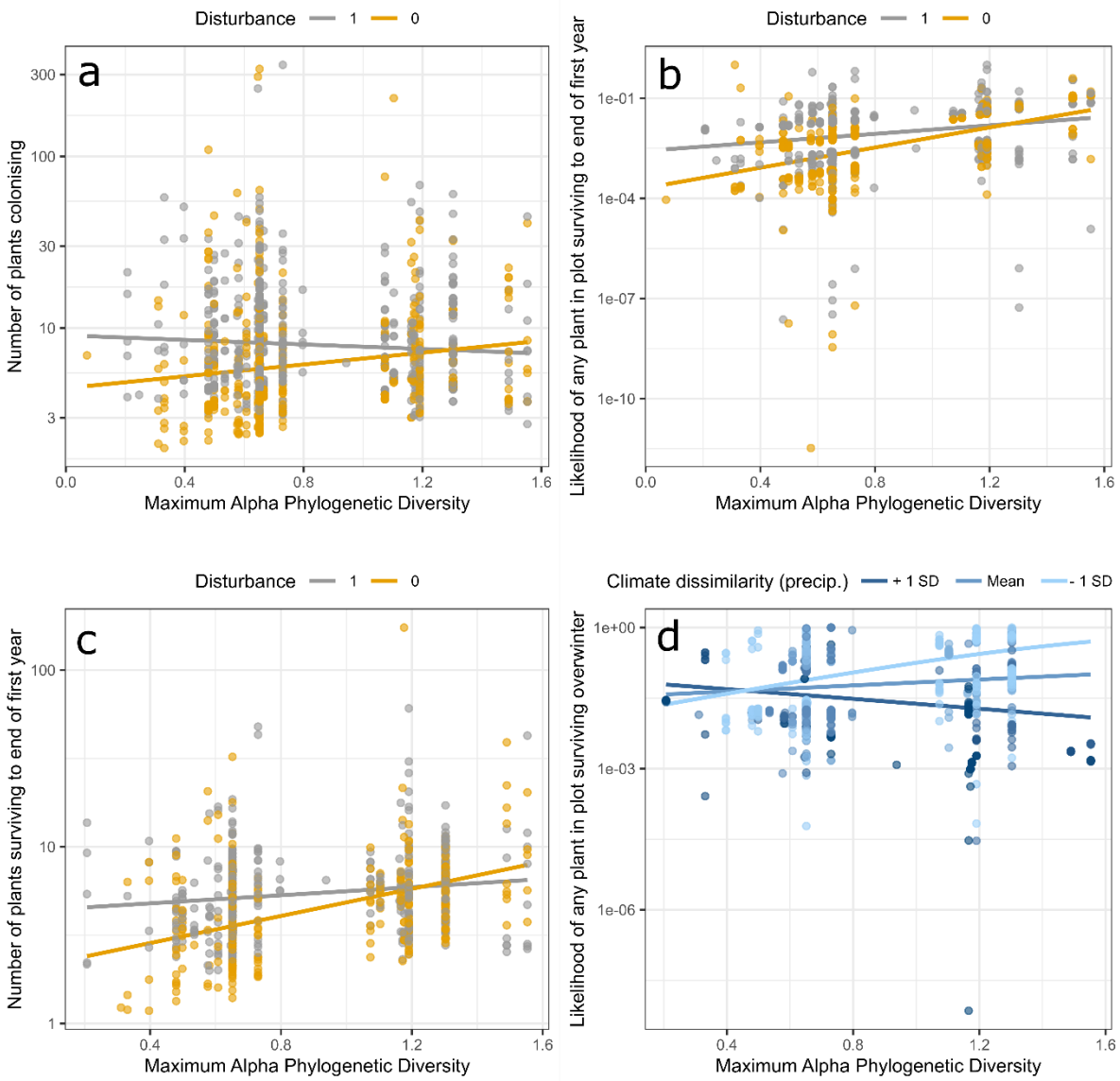


Figure S9: Marginal effects plot showing interactions between maximum alpha PD and disturbance on metrics of seeded species success. Plotted points represent observed data with all other fixed and random effects accounted for.

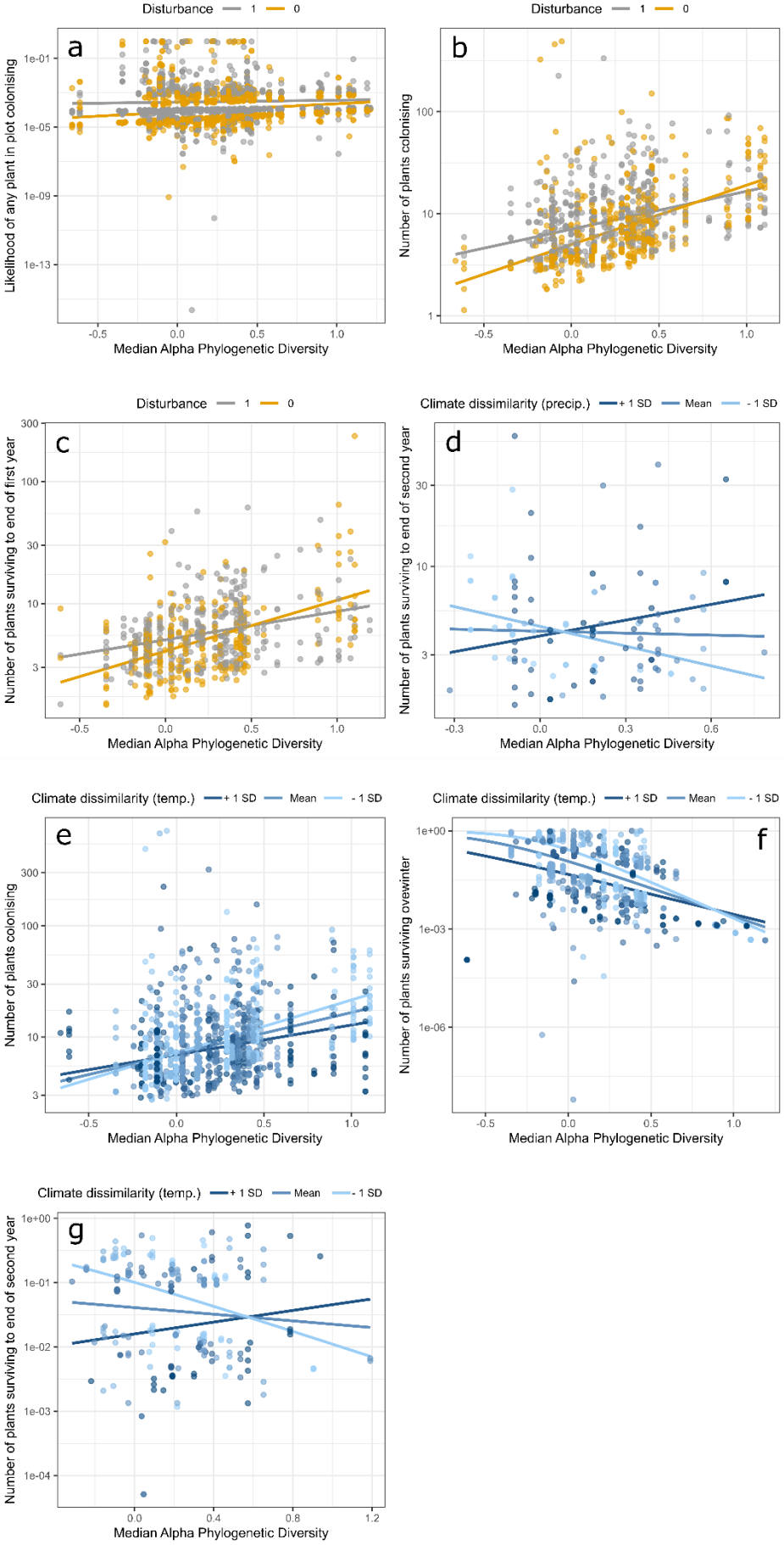
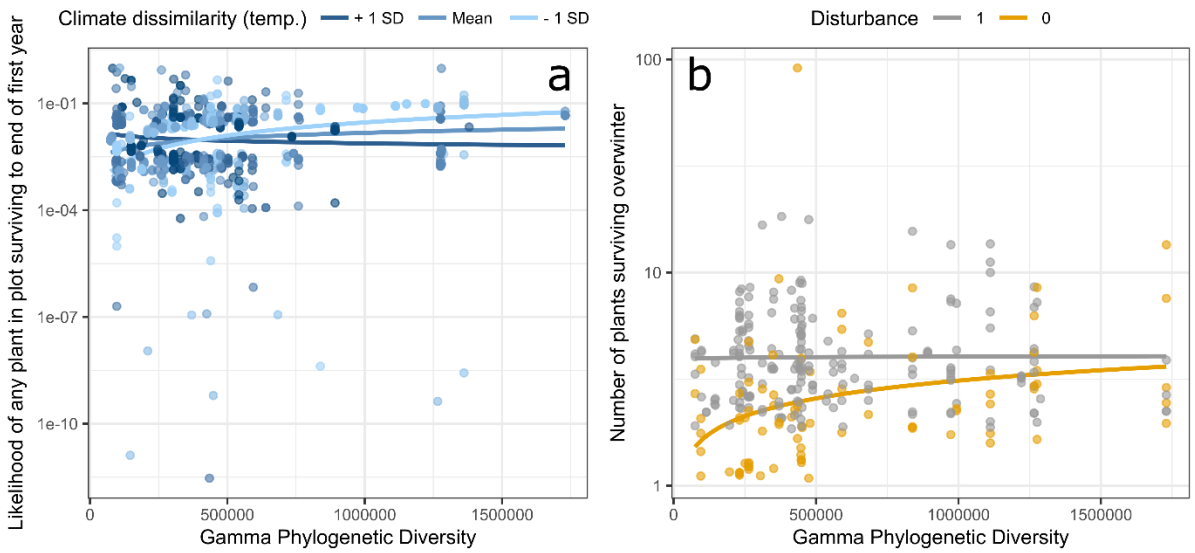

Disturbance 1 0
a
Likelihood of any plant in plot colonising
1e-01
1e-05
1e-09
1e-13
-0.5 0.0 0.5 1.0
Median Alpha Phylogenetic Diversity
Disturbance 1 0
b
Number of plants colonising
100
10
1
-0.5 0.0 0.5 1.0
Median Alpha Phylogenetic Diversity
Disturbance 1 0
c
Number of plants surviving to end of first year
300
100
30
10
3
-0.5 0.0 0.5 1.0
Median Alpha Phylogenetic Diversity
Climate dissimilarity (precip.) + 1 SD Mean - 1 SD
d
Number of plants surviving to end of second year
30
10
3
-0.3 0.0 0.3 0.6
Median Alpha Phylogenetic Diversity
Climate dissimilarity (temp.) + 1 SD Mean - 1 SD
e
Number of plants colonising
300
100
30
10
3
-0.5 0.0 0.5 1.0
Median Alpha Phylogenetic Diversity
Climate dissimilarity (temp.) + 1 SD Mean - 1 SD
f
Number of plants surviving ovewinter
1e+00
1e-03
1e-06
-0.5 0.0 0.5 1.0
Median Alpha Phylogenetic Diversity
Climate dissimilarity (temp.) + 1 SD Mean - 1 SD
g
Number of plants surviving to end of second year
1e+00
1e-01
1e-02
1e-03
1e-04
0.0 0.4 0.8 1.2
Median Alpha Phylogenetic Diversity

Figure S10 (previous page): Marginal effects plot showing interaction between median alpha PD, disturbance and climate dissimilarity on metrics of seeded species success. Plotted points represent observed data with all other fixed and random effects accounted for.



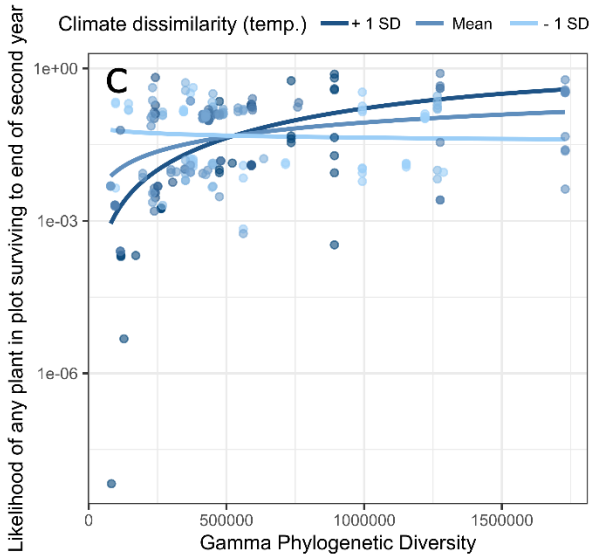


Figure S11: Marginal effects plot showing interaction between gamma PD, disturbance and temperature dissimilarity on metrics of seeded species success. Plotted points represent observed data with all other fixed and random effects accounted for.

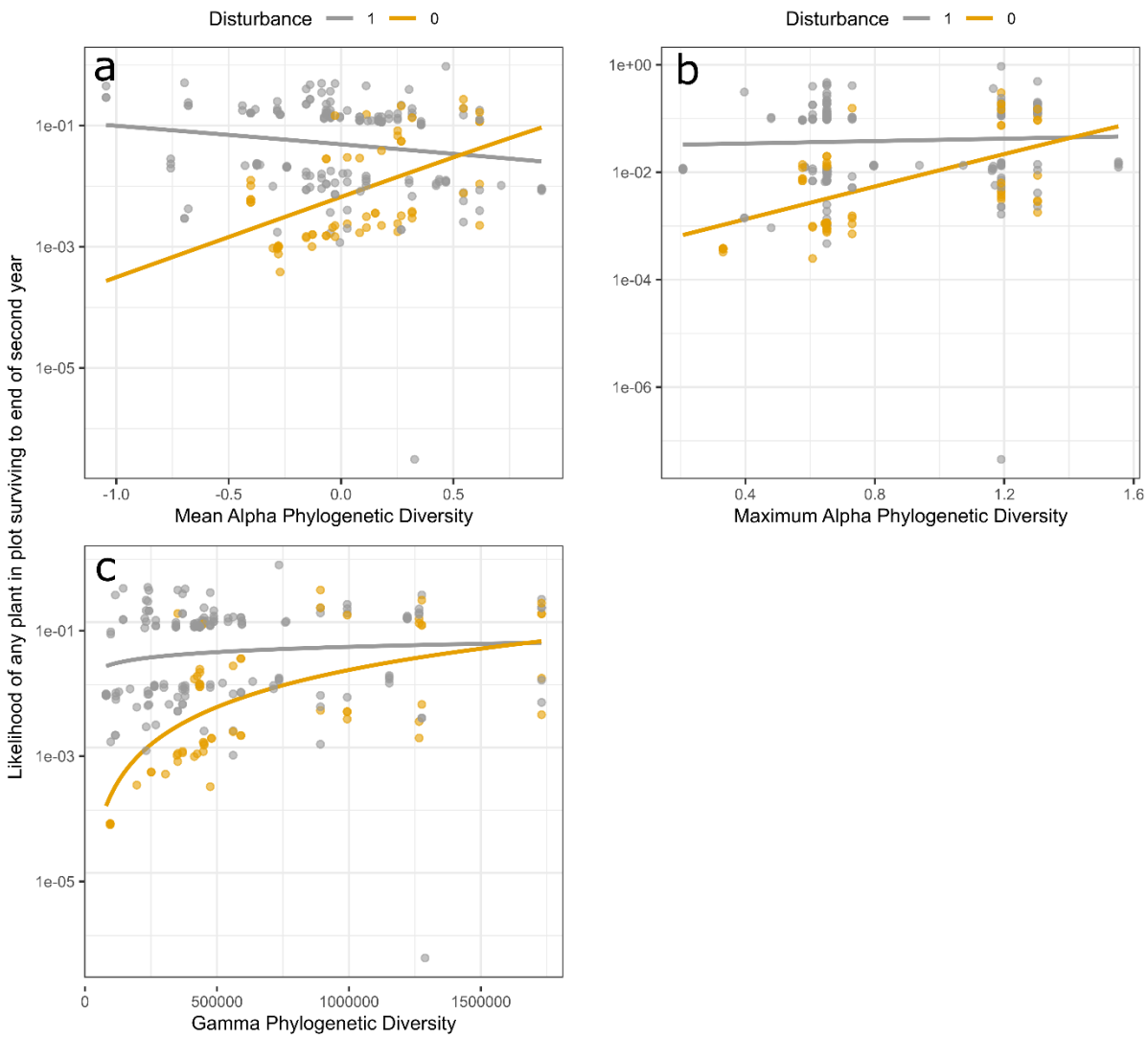


Figure S12: Marginal effects plot showing interaction between disturbance and (a) mean, (b) maximum and (c) gamma PD on the likelihood of species surviving to the end of the second growing season. These results are from the 'SC' models. Plotted points represent observed data with all other fixed and random effects accounted for.

Effect of relative indigenous-PD

Interactions between relative indigenous-PD and disturbance were also seen when using maximum alpha PD difference (Fig. S13a, c), median PD difference (Fig. S14d) or gamma PD difference (Fig. S15a), and all these interactions were consistent when using the 'SC' and 'Hnd' models (Tables S6-S9). Note that the figures make this trend difficult to see due to needing to correctly scale the y-axis, but in all cases the rate of increase of survival with higher relative indigenous-PD is higher in undisturbed rather than disturbed plots, supporting the trend. This interaction was also seen with median PD difference in the first year, but in the opposite direction (i.e. higher median PD difference only increases survival likelihood in undisturbed plots), potentially because surviving plants had skewed median alpha PD distributions (Fig. S14a, b).

The positive interaction between mean alpha PD difference and temperature dissimilarity was also seen in median PD for both the Haeuser-only and 'K+H' datasets (Figs. S14c, S16c); in other words, high relative indigenous-PD increased survival in the second year when the temperature dissimilarity was higher. The additional models also found interactions with precipitation dissimilarity: higher relative indigenous-PD only helped the *likelihood* of survival when precipitation dissimilarity was *low* (Figs. S15b; S16a, b) but only helped the *number* of plants surviving when precipitation dissimilarity was *high* (Figs. S13b, d; S14e). These effects were largely maintained in the 'SC' models (Tables S7 – S9). In addition, the trend of plants producing a higher number of flowers in more dissimilar precipitation regimes was consistent in the SC models (Tables S6, S7, S9). Cumulatively, these results suggest the same interpretation as in Question 1. Plants need to be similar enough ecologically for high relative indigenous-PD to be beneficial, but for those that can survive, they are more likely to take advantage of high relative indigenous-PD if they are ecologically different.

The above conclusions rely on a slight difference in interpretation of the effects of precipitation dissimilarity and temperature dissimilarity. Specifically, the PD*precipitation dissimilarity interaction is always *negative* for *likelihood* of survival (regardless of growing season) and *positive* for *number of plants* surviving (regardless of growing season). As described above, we have interpreted this as evidence that plants must be similar enough to the local community to have a chance of surviving (and therefore benefit from high indigenous-PD) but are more likely to become abundant if they are slightly different. In

contrast, the PD*temperature dissimilarity interaction is *negative* for likelihood of survival in the *first year*, but *positive* for likelihood of survival in the *second year*. We have also interpreted this as evidence that plants must be similar enough to the local community to have a chance of surviving (and therefore benefit from high indigenous-PD) but are more likely to become abundant if they are slightly different. We suggest that this is because precipitation dissimilarity overall provides a stronger selective pressure than temperature dissimilarity, especially into the second growing season (Figs. 2 and 4, main text) and so imposes a more sustained effect on likelihood of any plants in a plot surviving, as opposed to temperature dissimilarity whose significant negative effects are limited to the first growing season (Fig. 2 main text).

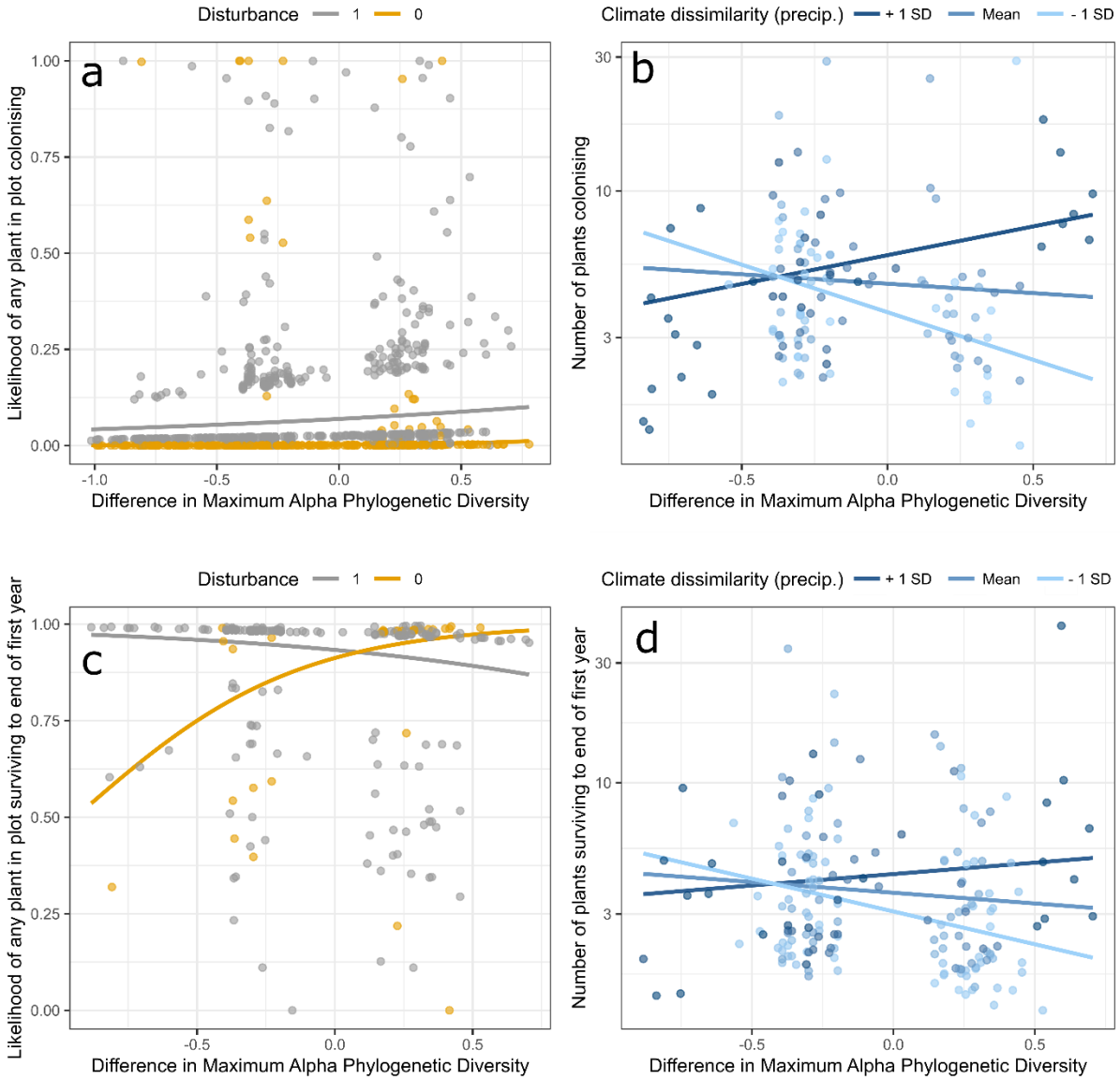


Figure S13: Marginal effects plot showing interaction between maximum alpha PD difference and climate dissimilarity or disturbance on plant responses at various time points (Haeuser data). Plotted points represent observed data with all other fixed and random effects accounted for. Note in (a) that the PD*disturbance interaction direction is not obvious, however the rate of increase with higher PD is faster in undisturbed than disturbed plots, so the trend is consistent with that of Figs. 3 and 5 (main text).

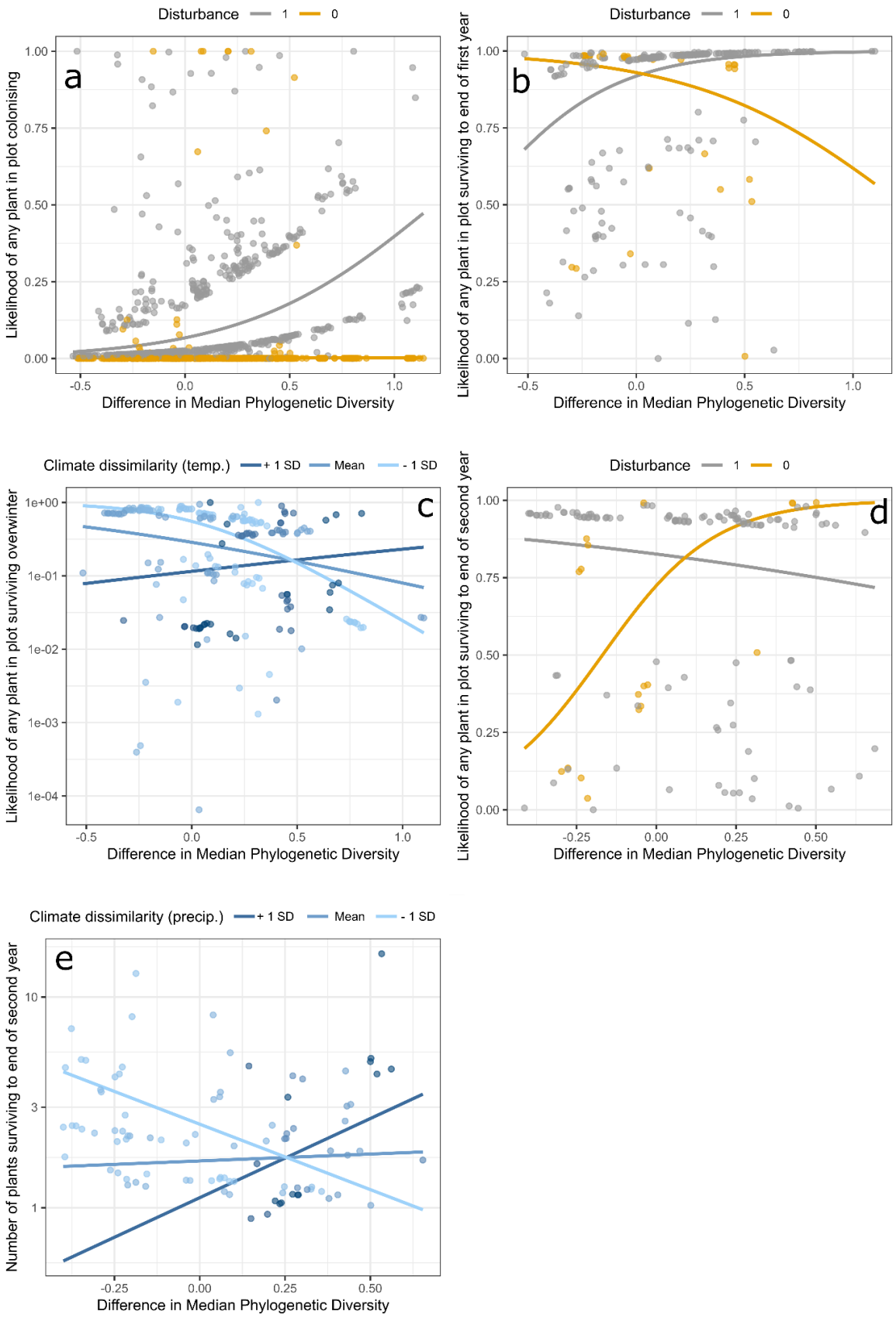


Figure S14: Marginal effects plot showing interaction between median alpha PD and climate dissimilarity or disturbance on plant responses at various time points (Haeuser data). Plotted points represent observed data with all other fixed and random effects accounted for.

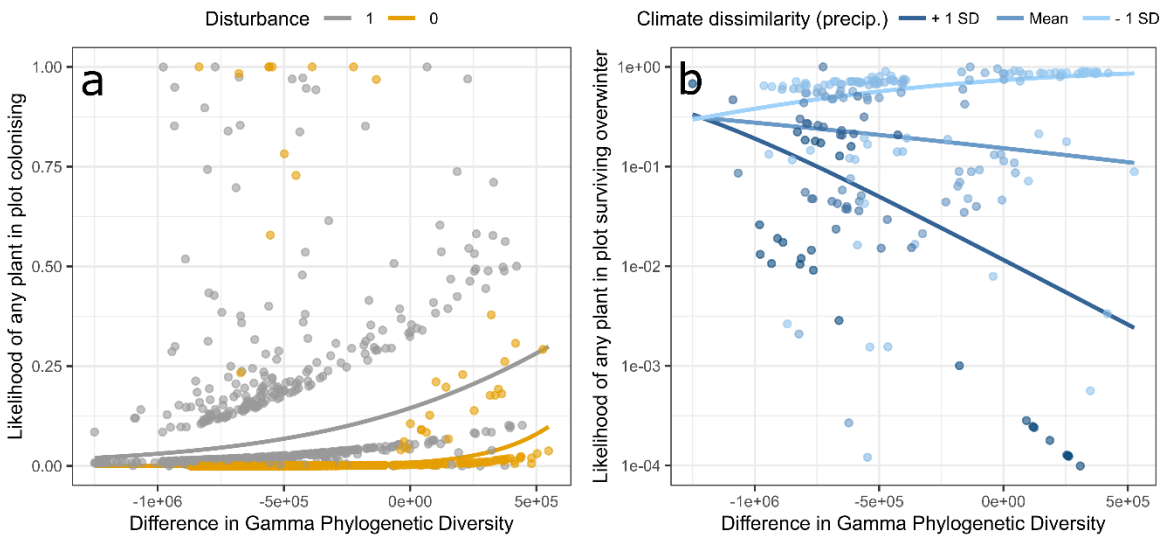


Figure S15: Marginal effects plot showing interaction between gamma alpha PD and climate dissimilarity or disturbance on plant responses at various time points (Haeuser data). Plotted points represent observed data with all other fixed and random effects accounted for. Note in (a) that the PD*disturbance interaction direction is not obvious, however the rate of increase with higher PD is faster in undisturbed than disturbed plots, so the trend is consistent with that of Figs. 2 and 4 (main text).

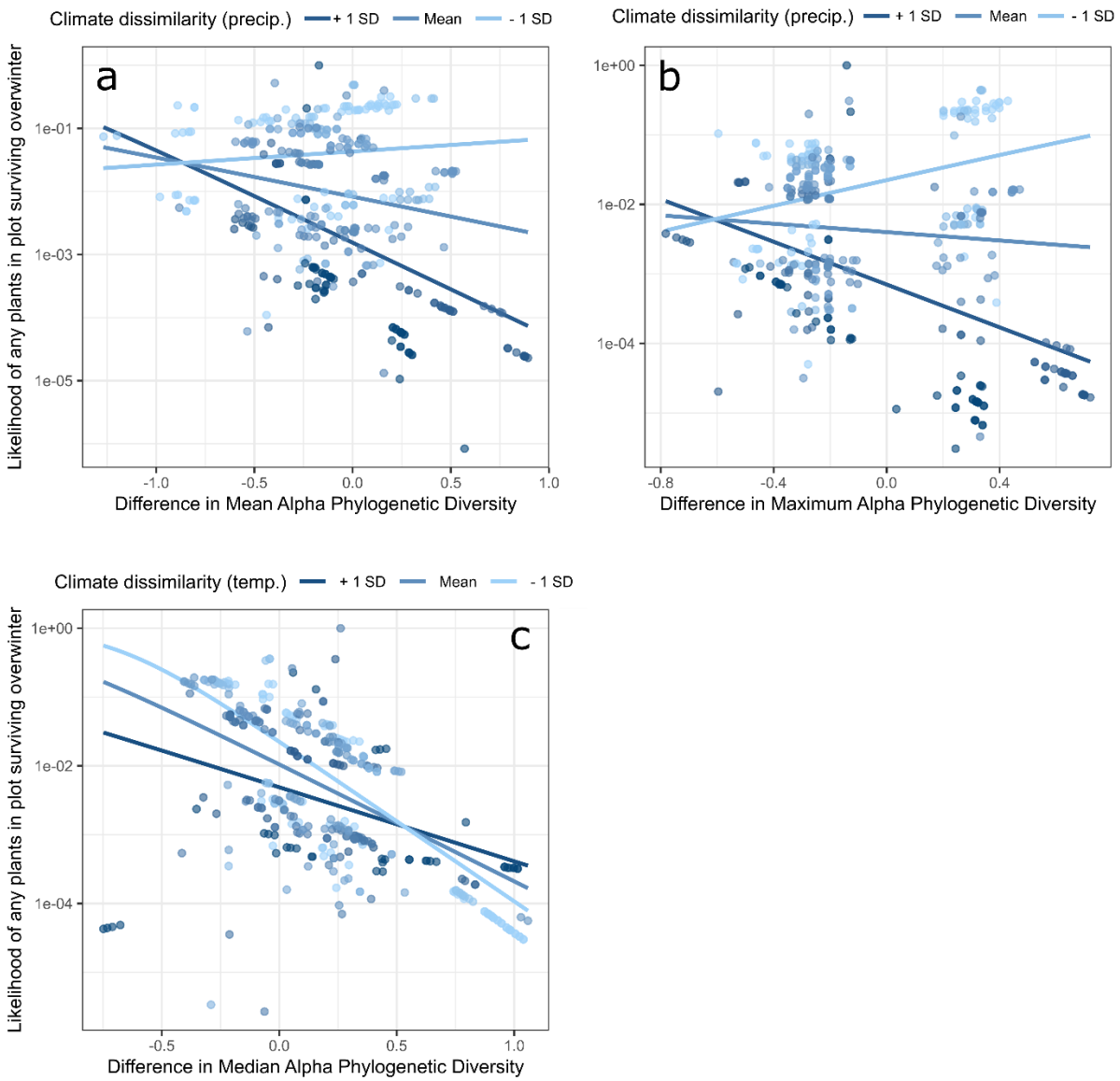


Figure S16: Marginal effects plot showing interaction between climate dissimilarity and (a) mean, (b) maximum and (c) median PD on the likelihood of overwinter plant survival, when using the simpler K+H difference in PD metric (see *Supplementary analyses: Robustness tests*). Plotted points represent observed data with all other fixed and random effects accounted for.

Effect of precipitation dissimilarity

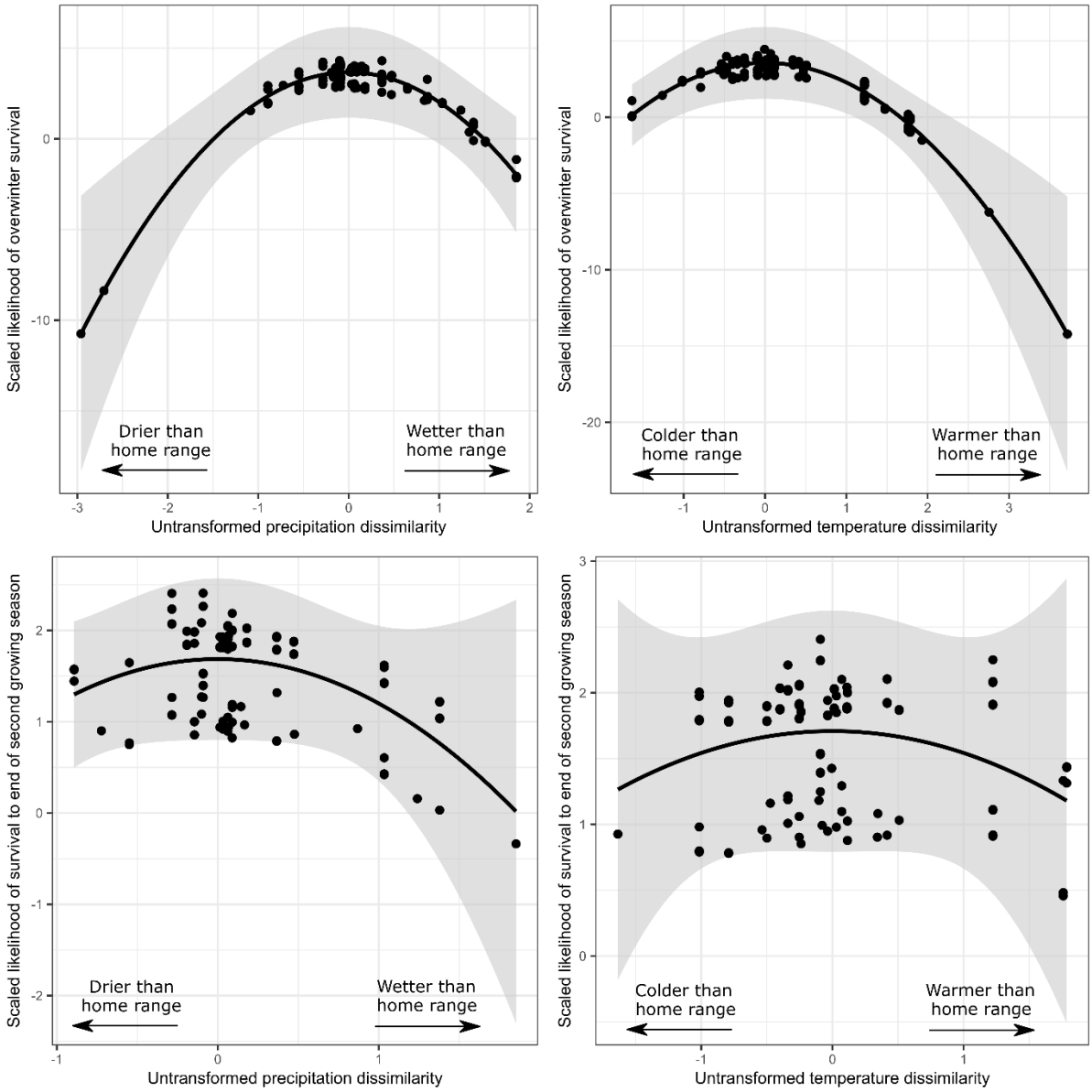


Figure S17: The negative relationship between precipitation dissimilarity or temperature dissimilarity and likelihood of overwinter and second-year survival is quadratic: species that came from increasingly wetter or drier climates than the experimental climate experienced lower success, and species that came from increasingly colder or warmer climates experienced lower success. Note that the absolute value of dissimilarity is used in the main model; this plot verifies that higher untransformed dissimilarity in both directions leads to lower survival.

Species origin and trait inclusion

In all cases, results held for both native and non-native species, and for species from different biogeographic regions (Table S10). Adding either of these variables did not add species origin added no information for 28 of 32 models (AIC within 2 of main models; species origin p-value > 0.05; Table S10); in the four cases where species origin was significant, the effect and significance of indigenous-PD was unchanged, suggesting that high indigenous-PD provides a benefit for establishment of both native and non-native species, and for non-native species regardless of specific continental origin and is therefore a general force contributing to community assembly and structure. There was one additional case where biogeographic origin was not significant, but its inclusion removed a significant PD*disturbance interaction (Table S10); in this case, the main effect of indigenous-PD remained positive and significant.

Models where the effect of indigenous-PD was mediated entirely by species traits were consistently inferior to those where indigenous-PD was allowed to have a direct effect (Table S11). Further, when including traits, indigenous-PD was still a significant positive predictor (where sample size allowed). While the included traits were also related to species success in the models, particularly seed mass, these results suggest that the positive effect of indigenous-PD detected in our study cannot be explained purely by SLA, plant height and seed mass. This indicates that indigenous-PD provides a holistic measure of fitness with value in its own right. We do note that in the Question 2 models, the significant effect of indigenous-PD was sometimes lost (Table S11). However, this was due to the significantly lower sample size, necessitated by only including species for which there was appropriate trait data. We found that the significance of indigenous-PD was lost even when not including trait data (i.e. an identical model to the main model but only using the species which had trait data), confirming that the lost significance was due to sample size and not trait inclusion. However, even in these cases, the models including a direct indigenous-PD effect were still substantially preferred to models not including this effect (over 2 AIC lower), strongly suggesting that having a direct effect of indigenous-PD is still substantially preferred to modelling indigenous-PD only though traits, even if indigenous-PD does not reach the $p<0.05$ threshold.

Supplementary tables for effect of absolute indigenous-PD

Table S2: Summary of results, using *mean alpha phylogenetic diversity* as response variable. Bold indicates significant p-value. In the p-value column, the superscript K+M indicates the significance (either significant or not significant) was the same in the reduced model just using the Kempel and Muller datasets; the superscript K+H indicates the significance was the same in the reduced model using the Kempel and Haeuser datasets using the (modified) difference in phylogenetic diversity; the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components. Note that the K+H models did not include herbivory (indicated by NA), as only the Müller dataset had a closed herbivory treatment. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	**Stage**	**Explanatory variable**	**Effect size**	**p-value**
Any plants in plot	Colonisation	Mean alpha PD	-0.281 (-0.231)	0.522 $^{K+M; K+H; SC}$
		Disturbed	1.239	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	0.038 (0.060)	0.897 $^{K+M; K+H; SC}$
		Climate diss. (temp.)	0.037 (0.038)	0.916 $^{K+M; K+H; SC}$
	First growing season	Mean alpha PD	2.244 (1.881)	**0.006** $^{K+M; K+H; SC}$
		Disturbed	1.047	**0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-2.502 (-3.774)	**<0.001** $^{K+M; K+H}$
		Climate diss. (temp.)	0.712 (0.679)	0.275 $^{K+M; K+H; SC}$
		Herbivory	-0.628	0.096 $^{K+M; NA; SC}$
	Overwinter	Mean alpha PD	-0.332 (-0.299)	0.586 $^{K+M; K+H; SC}$
		Disturbed	1.895	**<0.001** $^{K+M; SC}$
		Climate diss. (precip.)	-1.184 (-1.665)	**0.005** $^{K+M; K+H; SC}$
		Climate diss. (temp.)	-0.978 (-0.888)	0.098 $^{K+M; K+H; SC}$
		Herbivory	-0.960	**0.001** $^{K+M; NA; SC}$
	Second growing season	Mean alpha PD	-1.431 (0.391)	0.186 $^{K+M; K+H; SC}$ (0.489)
		Disturbed	1.372	**0.028** $^{K+H; SC}$
		Climate diss. (precip.)	-2.313 (-2.724)	**0.001** $^{K+M; K+H; SC}$
		Climate diss. (temp.)	-1.026 (-0.995)	0.138 $^{K+M; K+H; SC}$ (0.152)
		Mean alpha PD*Climate diss. (temp.)	3.249 (2.385)	**0.013** $^{K+H}$
Number of plants in plot	Colonisation	Mean alpha PD	0.339 (0.283)	0.120 $^{K+M; K+H; SC}$
		Disturbed	0.228 (0.248)	**0.004** $^{K+M; K+H; SC}$ (**0.001**)

		Climate diss. (precip.)	0.226 (0.337)	0.072 K+M; SC
		Climate diss. (temp.)	-0.355 (-0.317)	**0.032** K+H
		Mean alpha PD* Disturbed	-0.271 (-0.226)	**0.033** K+M; K+H; SC
	First growing season	Mean alpha PD	0.588 (0.536)	**0.004** K+M; K+H; SC
		Disturbed	0.153 (0.159)	**0.026** K+M; SC (**0.020**)
		Climate diss. (precip.)	-0.001 (<0.001)	0.999 K+M; K+H; SC
		Climate diss. (temp.)	-0.361 (-0.332)	**0.018**
		Herbivory	-0.091	0.198 K+M; NA; SC
		Mean alpha PD* Disturbed	-0.237 (-0.216)	**0.047** K+M; K+H; SC
	Overwinter	Mean alpha PD	0.244 (0.197)	0.280 K+M; K+H; SC
		Disturbed	0.464	**0.001** K+M; SC
		Climate diss. (precip.)	0.234 (0.293)	0.180 K+M; K+H; SC
		Climate diss. (temp.)	-0.214 (-0.135)	0.490 K+M; K+H; SC
		Herbivory	-0.030	0.800 K+M; NA; SC
	Second growing season	Mean alpha PD	0.174 (0.123)	0.530 K+H; SC
		Disturbed	0.076	0.688 K+M; K+H; SC
		Climate diss. (precip.)	0.442 (0.362)	0.162 K+M; K+H; SC
		Climate diss. (temp.)	0.017 (0.012)	0.956 K+M, K+H; SC

Table S3: Summary of results, using *maximum alpha phylogenetic diversity* as response variable. Bold indicates significant p-value. In the p-value column, the superscript K+M indicates the significance (either significant or not significant) was the same in the reduced model just using the Kempel and Muller datasets; the superscript K+H indicates the significance was the same in the reduced model using the Kempel and Haeuser datasets using the (modified) difference in phylogenetic diversity; the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components.. Note that the K+H models did not include herbivory (indicated by NA), as only the Müller dataset had a closed herbivory treatment. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Maximum alpha PD	-0.134 (-0.084)	0.814 $^{K+M;\ K+H;\ SC}$
		Disturbed	1.239	**<0.001** $^{K+M;\ K+H;\ SC}$
		Climate diss. (precip.)	0.022 (0.034)	0.940 $^{K+M;\ K+H;\ SC}$
		Climate diss. (temp.)	0.065 (0.067)	0.853 $^{K+M;\ K+H;\ SC}$
	First growing season	Maximum alpha PD	3.487 (2.268)	**0.001** ^{SC}
		Disturbed	2.559 (0.839)	**<0.001** $^{K+M;\ K+H;\ SC}$ (**0.013**)
		Climate diss. (precip.)	-2.094 (-3.159)	**0.001** $^{K+M;\ K+H}$
		Climate diss. (temp.)	0.499 (0.476)	0.388 $^{K+M;\ K+H;\ SC}$
		Herbivory	-0.646	0.092 $^{K+M;\ NA;\ SC}$
		Maximum alpha PD * Disturbed	-2.012 (-1.309)	**0.005** $^{K+M;\ SC}$
	Overwinter	Maximum alpha PD	2.882 (0.514)	**0.022** $^{K+H;\ SC}$ (0.347)
		Disturbed	1.912	**<0.001** $^{K+M;\ SC}$
		Climate diss. (precip.)	1.308 (-1.756)	0.193 $^{K+M}$ (**0.004**)
		Climate diss. (temp.)	-1.008 (-0.918)	0.092 $^{K+M;\ K+H}$
		Herbivory	-0.973	**0.006** $^{K+M;\ NA;\ SC}$
		Maximum alpha PD*Climate diss. (precip.)	-2.859 (-2.665)	**0.008** $^{K+M;\ K+H}$
	Second growing season	Maximum alpha PD	0.777 (0.496)	0.405 $^{K+M;\ K+H;\ SC}$
		Disturbed	1.327	**0.029** $^{K+H;\ SC}$
		Climate diss. (precip.)	-2.183 (-2.571)	**0.001** $^{K+H;\ SC}$
		Climate diss. (temp.)	-0.664 (-0.673)	0.291 $^{K+M;\ K+H;\ SC}$
Number of plants in plot	Colonisation	Maximum alpha PD	0.397 (0.247)	0.141 $^{K+M;\ SC}$
		Disturbed	0.708 (0.262)	**<0.001** $^{K+M;\ K+H;\ SC}$
		Climate diss. (precip.)	0.226 (0.336)	0.072 $^{;\ SC}$

		Climate diss. (temp.)	-0.385 (-0.343)	**0.016** K+H
		Maximum alpha PD * Disturbed	-0.549 (-0.342)	**<0.001** K+M; K+H; SC
	First growing season	Maximum alpha PD	0.884 (0.587)	**<0.001** K+M; K+H; SC
		Disturbed	0.766 (0.205)	**<0.001** K+M; SC (**0.003**)
		Climate diss. (precip.)	-0.013 (-0.018)	0.918 K+M; K+H; SC
		Climate diss. (temp.)	-0.401 (-0.369)	**0.007** K+H
		Herbivory	-0.090	0.201 K+M; NA; SC
		Maximum alpha PD* Disturbed	-0.620 (-0.412)	**<0.001** K+M; K+H; SC
	Overwinter	Maximum alpha PD	-0.051 (-0.031)	0.867 K+M; K+H; SC
		Disturbed	0.451	**0.001** K+M; SC
		Climate diss. (precip.)	0.218 (0.273)	0.213 K+M; K+H; SC
		Climate diss. (temp.)	-0.213 (-0.134)	0.497 K+M; K+H; SC
		Herbivory	-0.025	0.832 K+M; NA; SC
	Second growing season	Maximum alpha PD	-0.181 (-0.107)	0.594 K+M; K+H; SC
		Disturbed	0.038	0.845 K+M; K+H; SC
		Climate diss. (precip.)	0.416 (0.340)	0.195 K+M; K+H; SC
		Climate diss. (temp.)	0.050 (0.035)	0.871 K+M; K+H; SC

Table S4: Summary of results, using *median alpha phylogenetic diversity* as response variable. Bold indicates significant p-value. In the p-value column, the superscript K+M indicates the significance (either significant or not significant) was the same in the reduced model just using the Kempel and Muller datasets; the superscript K+H indicates the significance was the same in the reduced model using the Kempel and Haeuser datasets using the (modified) difference in phylogenetic diversity; the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components.. Note that the K+H models did not include herbivory (indicated by NA), as only the Müller dataset had a closed herbivory treatment. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Median alpha PD	1.105 (0.715)	0.072 $^{K+M; K+H; SC}$
		Disturbed	1.317 (-1.151)	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	0.091 (-0.142)	0.767 $^{K+M; K+H; SC}$
		Climate diss. (temp.)	0.201 (0.208)	0.579 $^{K+M; K+H; SC}$
		Median alpha PD*Disturbed	-0.839 (-0.543)	**0.016** $^{K+M; SC}$
	First growing season	Median alpha PD	2.417 (1.623)	**0.023** $^{K+M; K+H; SC}$
		Disturbed	1.018	**0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-2.832 (-4.272)	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (temp.)	0.947 (0.904)	0.168 $^{K+M; SC}$
		Herbivory	-0.630	0.094 $^{K+M; NA; SC}$
	Overwinter	Median alpha PD	-5.103 (-2.666)	**<0.001** $^{K+H; SC}$
		Disturbed	1.997	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-0.401 (-0.564)	0.319 $^{K+M}$
		Climate diss. (temp.)	-2.179 (1.519)	**0.002** $^{K+M; K+H}$
		Herbivory	-0.944	**0.001** $^{K+M; NA; SC}$
		Median alpha PD * Climate diss. (temp.)	2.480 (1.513)	**0.028** $^{K+H}$
	Second growing season	Median alpha PD	-2.642 (-0.343)	0.059 $^{K+M; K+H; SC}$ (0.588)
		Disturbed	1.426	**0.021** $^{K+H; SC}$
		Climate diss. (precip.)	-1.977 (-2.329)	**0.002** $^{K+H; SC}$
		Climate diss. (temp.)	-1.911 (-1.171)	**0.049** (0.108)
		Median alpha PD * Climate diss. (temp.)	3.342 (1.889)	**0.043** $^{K+H}$
Number of plants in plot	Colonisation	Median alpha PD	1.590 (0.848)	**<0.001** $^{K+M; K+H; SC}$
		Disturbed	0.346 (0.238)	**<0.001** $^{K+M; K+H; SC}$ (**0.002**)

		Climate diss. (precip.)	0.058 (0.086)	0.647 [K+M; SC]
		Climate diss. (temp.)	-0.045 (-0.147)	0.792 [K+M; K+H; SC] (0.309)
		Median alpha PD*Disturbed	-0.482 (-0.306)	**0.002** [K+M; K+H; SC]
		Median alpha PD * Climate diss. (temp.)	-0.537 (-0.303)	**0.033** [K+H]
	First growing season	Median alpha PD	0.960 (0.650)	**<0.001** [K+M, K+H; SC]
		Disturbed	0.213 (0.123)	**0.002** [K+M; SC] (0.080)
		Climate diss. (precip.)	-0.097 (-0.135)	0.495 [K+M; K+H; SC]
		Climate diss. (temp.)	-0.253 (-0.233)	0.100 [K+M; K+H; SC]
		Herbivory	-0.090	0.204 [K+M; NA; SC]
		Median alpha PD* Disturbed	-0.425 (-0.287)	**0.012** [K+M; K+H; SC]
	Overwinter	Median alpha PD	0.299 (0.138)	0.410 [K+M; SC]
		Disturbed	0.449	**0.001** [K+M; SC]
		Climate diss. (precip.)	0.200 (0.250)	0.253 [K+M; K+H; SC]
		Climate diss. (temp.)	-0.216 (-0.135)	0.483 [K+M; SC]
		Herbivory	-0.029	0.802 [K+M; NA; SC]
	Second growing season	Median alpha PD	-1.166 (-0.044)	0.064 [K+H, K+H; SC] (0.820)
		Disturbed	-0.016	0.935 [K+M; K+H; SC]
		Climate diss. (precip.)	-0.158 (0.135)	0.724 [K+M; K+H; SC] (0.635)
		Climate diss. (temp.)	0.239 (0.168)	0.439 [K+M; K+H; SC]
		Median alpha PD * Climate diss. (temp.)	1.964 (0.795)	**0.042** [K+H; SC]

Table S5: Summary of results, using *gamma phylogenetic diversity* as response variable. Bold indicates significant p-value. In the p-value column, the superscript K+M indicates the significance (either significant or not significant) was the same in the reduced model just using the Kempel and Muller datasets; the superscript K+H indicates the significance was the same in the reduced model using the Kempel and Haeuser datasets using the (modified) difference in phylogenetic diversity; the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components.. Note that the K+H models did not include herbivory (indicated by NA), as only the Müller dataset had a closed herbivory treatment. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Gamma PD	-0.329 (-0.464)	0.192 $^{K+M, K+H; SC}$
		Disturbed	1.238	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-0.072 (-0.113)	0.808 $^{K+M; K+H; SC}$
		Climate diss. (temp.)	0.055 (0.057)	0.873 $^{K+H; SC}$
	First growing season	Gamma PD	1.325 (0.733)	**0.034** $^{K+H; SC}$ (0.266)
		Disturbed	1.017	**0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-2.305 (-3.477)	**<0.001** $^{K+M; K+H}$
		Climate diss. (temp.)	20.085 (0.105)	**0.038** $^{K+H}$ (0.868)
		Herbivory	-0.616	0.102 $^{K+M; NA; SC}$
		Gamma PD * Climate diss. (temp.)	-1.558 (-2.169)	**0.042** $^{K+H}$
	Overwinter	Gamma PD	0.371 (0.560)	0.275 $^{K+M; K+H SC}$
		Disturbed	1.906	**<0.001** $^{K+M; K+H; SC}$
		Climate diss. (precip.)	-1.096 (-1.540)	**0.010** $^{K+H; SC}$
		Climate diss. (temp.)	-0.908 (-0.827)	0.115 $^{K+M; K+H; SC}$
		Herbivory	-0.970	**0.001** $^{K+M; NA; SC}$
	Second growing season	Gamma PD	-0.369 (1.488)	0.564 $^{K+M; K+H; SC}$ (**0.032**)
		Disturbed	1.488	**0.021** $^{K+H; SC}$
		Climate diss. (precip.)	-2.235 (-2.633)	**0.001** $^{K+H; SC}$
		Climate diss. (temp.)	-29.613 (-0.401)	**0.034** (0.571)
		Gamma PD * Climate diss. (temp.)	2.248 (3.411)	**0.035**
Number of plants in plot	Colonisation	Gamma PD	-0.091 (-0.139)	0.391 $^{K+H; SC}$
		Disturbed	0.260	**0.001** $^{K+M; SC}$
		Climate diss. (precip.)	0.213 (0.318)	0.087 ^{SC}

		Climate diss. (temp.)	-0.397 (-0.354)	**0.011** K+H
	First growing season	Gamma PD	0.144 (0.218)	0.155 K+M; K+H; SC
		Disturbed	0.175	**0.011** K+M; SC
		Climate diss. (precip.)	0.085 (0.118)	0.538 K+M; K+H; SC
		Climate diss. (temp.)	-0.425 (-0.392)	**0.005** K+M; K+H
		Herbivory	-0.090	0.207 K+M; NA; SC
	Overwinter	Gamma PD	0.277 (0.406)	0.098 K+M; K+H; SC
		Disturbed	3.974 (0.475)	**0.027** K+M; SC (**<0.001**)
		Climate diss. (precip.)	0.311 (0.389)	0.104 K+M; K+H; SC
		Climate diss. (temp.)	-0.245 (-0.154)	0.442 K+M; K+H; SC
		Herbivory	-0.028	0.811 K+M; NA; SC
		Gamma PD * Disturbed	-0.269 (-0.394)	**0.050** SC
	Second growing season	Gamma PD	0.095 (0.120)	0.546 K+M; K+H; SC
		Disturbed	-0.087	0.652 K+M; K+H; SC
		Climate diss. (precip.)	0.468 (0.383)	0.146 K+M; K+H; SC
		Climate diss. (temp.)	0.016 (0.011)	0.958 K+M; K+H; SC

Supplementary tables for effect of relative indigenous-PD

Table S6: Summary of results, using *difference in mean alpha phylogenetic diversity* (planted species PD minus abundance-weighted community PD) as response variable. Bold indicates significant p-value. In the p-value column, the superscript Hnd indicates the significance (either significant or not significant) was the same in a model using raw phylogenetic diversity (rather than the difference); the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Mean alpha PD diff.	1.881 (1.827)	**0.009** $^{Hnd; SC}$
		Disturbed	3.644	**<0.001** $^{Hnd; SC}$
		Climate diss. (precip.)	-0.909 (-1.273)	0.126 $^{Hnd; SC}$
		Climate diss. (temp.)	-0.183 (0.273)	0.743 $^{Hnd; SC}$
		Heating	-0.855	**0.004** $^{Hnd; SC}$
		awMPD	-0.325 (-0.650)	**0.045** ^{SC}
	First growing season	Mean alpha PD diff.	1.743 (1.471)	**0.005** $^{Hnd; SC}$
		Disturbed	0.409	0.467 $^{Hnd; SC}$
		Climate diss. (precip.)	-0.306 (-0.357)	0.522 $^{Hnd; SC}$
		Climate diss. (temp.)	0.542 (0.641)	0.279 $^{Hnd; SC}$
		Heating	-1.029	**0.006** $^{Hnd; SC}$
		awMPD	-0.045 (-0.091)	0.847 $^{Hnd; SC}$
	Overwinter	Mean alpha PD diff.	-2.460 (-0.825)	**0.050** (0.256)
		Disturbed	1.237	**0.047** ^{Hnd}
		Climate diss. (precip.)	-2.148 (2.403)	**<0.001** $^{Hnd; SC}$
		Climate diss. (temp.)	-0.692 (-1.275)	0.252 $^{Hnd;SC}$ (0.095)
		Heating	0.798	0.065 $^{Hnd; SC}$
		awMPD	-0.388 (-0.769)	0.167 $^{Hnd; SC}$
		Mean alpha PD diff. * Climate diss. (temp.)	2.896 (2.831)	**0.053** ^{Hnd}
	Second growing season	Mean alpha PD diff.	13.779 (10.292)	0.054 $^{Hnd; SC}$
		Disturbed	2.265 (4.624)	0.080 ^{Hnd} (0.052)
		Climate diss. (precip.)	-0.904 (-0.724)	0.198 $^{Hnd; SC}$
		Climate diss. (temp.)	0.012 (0.012)	0.983 $^{Hnd; SC}$
		Heating	-0.936	**0.033** ^{SC}
		awMPD	-0.100 (-0.199)	0.675 $^{Hnd; SC}$
		Mean alpha PD diff. * Disturbed	-13.859 (-10.352)	**0.052** ^{SC}

	Flowering*	Mean alpha PD diff.	0.986 (0.963)	0.305 [Hnd; SC]
		Climate diss. (precip.)	-0.689 (-0.965)	0.412 [Hnd; SC]
		Climate diss. (temp.)	-0.454 (-0.716)	0.614 [Hnd; SC]
		Heating	-0.649	0.072 [Hnd; SC]
		awMPD	0.011 (0.022)	0.967 [Hnd; SC]
Number of plants in plot	Colonisation	Mean alpha PD diff.	0.839 (0.642)	**0.006** [Hnd; SC]
		Disturbed	1.568	**<0.001** [Hnd; SC]
		Climate diss. (precip.)	0.113 (0.141)	0.579 [Hnd; SC]
		Climate diss. (temp.)	-0.287 (-0.327)	0.223 [Hnd; SC]
		Heating	-0.310	**0.013** [Hnd; SC]
		awMPD	0.057 (0.114)	0.321 [Hnd; SC]
	First growing season	Mean alpha PD diff.	0.597 (0.455)	**0.003** [Hnd; SC]
		Disturbed	-0.018	0.915 [Hnd; SC]
		Climate diss. (precip.)	0.122 (0.141)	0.397 [Hnd; SC]
		Climate diss. (temp.)	-0.219 (-0.274)	0.103 [Hnd; SC]
		Heating	-0.237	**0.037** [Hnd; SC]
		awMPD	-0.086 (-0.171)	0.209 [Hnd; SC]
	Overwinter	Mean alpha PD diff.	0.336 (0.243)	0.165 [Hnd; SC]
		Disturbed	0.151	0.533 [Hnd; SC]
		Climate diss. (precip.)	0.404 (0.334)	**0.044** [Hnd]
		Climate diss. (temp.)	-0.145 (-0.127)	0.471 [Hnd; SC]
		Heating	-0.139	0.262 [Hnd; SC]
		awMPD	0.111 (0.221)	0.157 [Hnd; SC]
	Second growing season	Mean alpha PD diff.	0.367 (0.272)	0.233 [Hnd; SC]
		Disturbed	0.084	0.773 [Hnd; SC]
		Climate diss. (precip.)	0.488 (0.381)	0.084 [Hnd; SC]
		Climate diss. (temp.)	-0.163 (-0.152)	0.512 [Hnd; SC]
		Heating	-0.051	0.735 [Hnd; SC]
		awMPD	0.018 (0.035)	0.861 [Hnd; SC]
	Flowering*	Mean alpha PD diff.	-0.145 (-0.139)	0.481 [Hnd; SC]
		Climate diss. (precip.)	0.357 (0.432)	**0.028** [Hnd; SC]
		Climate diss. (temp.)	-0.306 (-0.355)	0.100 [Hnd; SC]
		Heating	0.081	0.518 [Hnd; SC]
		awMPD	-0.096 (-0.191)	0.237 [Hnd; SC]

*Plants only flowered in disturbed plots, so these models exclude disturbance

Table S7: Summary of results, using *difference in maximum alpha phylogenetic diversity* (planted species PD minus abundance-weighted community PD) as response variable. Bold indicates significant p-value. In the p-value column, the superscript Hnd indicates the significance (either significant or not significant) was the same in a model using raw phylogenetic diversity (rather than the difference); the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Maximum alpha PD diff.	2.298 (1.742)	**0.046** $^{Hnd; SC}$
		Disturbed	3.649 (3.848)	**<0.001** $^{Hnd; SC}$
		Climate diss. (precip.)	-0.817 (-1.144)	0.199 $^{Hnd; SC}$
		Climate diss. (temp.)	0.320 (0.476)	0.588 $^{Hnd; SC}$
		Heating	-0.874	**0.009** $^{Hnd; SC}$
		awMPD	-0.012 (-0.522)	0.112 $^{Hnd; SC}$
		Maximum alpha PD diff. * Disturbed	-1.776 (-1.347)	**0.034** $^{Hnd; SC}$
	First growing season	Maximum alpha PD diff.	2.488 (1.748)	0.165 $^{Hnd; SC}$
		Disturbed	0.296 (0.445)	0.623 ^{SC} (0.460)
		Climate diss. (precip.)	-0.406 (-0.474)	0.465 $^{Hnd; SC}$
		Climate diss. (temp.)	0.632 (0.748)	0.261 $^{Hnd; SC}$
		Heating	-1.115	**0.004** $^{Hnd; SC}$
		awMPD	-0.001 (-0.003)	0.995 $^{Hnd; SC}$
		Maximum alpha PD diff.* Disturbed	-3.542 (-2.488)	**0.041** $^{Hnd; SC}$
	Overwinter	Maximum alpha PD diff.	-0.786 (-0.549)	0.419 $^{Hnd; SC}$
		Disturbed	1.192	0.058 $^{Hnd; SC}$
		Climate diss. (precip.)	-2.485 (-2.780)	**0.001** $^{Hnd; SC}$
		Climate diss. (temp.)	-0.825 (0.969)	0.190 $^{Hnd; SC}$
		Heating	0.845	0.051 $^{Hnd; SC}$
		awMPD	-0.379 (-0.758)	0.181 $^{Hnd; SC}$
	Second growing season	Maximum alpha PD diff.	0.188 (0.117)	0.824 $^{Hnd; SC}$
		Disturbed	0.727	0.259 $^{Hnd; SC}$
		Climate diss. (precip.)	-0.672 (-0.538)	0.325 $^{Hnd; SC}$
		Climate diss. (temp.)	0.135 (0.129)	0.789 $^{Hnd; SC}$
		Heating	-0.756	0.068 $^{Hnd; SC}$
		awMPD	-0.059 (-0.118)	0.799 $^{Hnd; SC}$
	Flowering*	Maximum alpha PD diff.	0.168 (0.127)	0.889 $^{Hnd; SC}$

		Climate diss. (precip.)	-0.666 (-0.933)	0.443 [Hnd; SC]
		Climate diss. (temp.)	-0.468 (-0.696)	0.569 [Hnd; SC]
		Heating	-0.701	0.053 [Hnd; SC]
		awMPD	0.021 (0.042)	0.938 [Hnd; SC]
Number of plants in plot	Colonisation	Maximum alpha PD diff.	-0.865 (-0.613)	0.102 [Hnd; SC]
		Disturbed	1.545	**<0.001** [Hnd; SC]
		Climate diss. (precip.)	0.375 (0.246)	0.113 [Hnd; SC] (0.276)
		Climate diss. (temp.)	-0.339 (-0.441)	0.883 [Hnd; SC]
		Heating	-0.356	**0.002** [Hnd; SC]
		awMPD	0.050 (0.100)	0.384 [Hnd; SC]
		Maximum alpha PD diff.*Climate diss. (precip.)	1.002 (0.710)	**0.022** [Hnd; SC]
	First growing season	Maximum alpha PD diff.	-0.565 (-0.139)	0.092 [Hnd; SC] (0.414)
		Disturbed	-0.074	0.672 [Hnd; SC]
		Climate diss. (precip.)	0.297 (0.308)	0.055 [Hnd; SC] (0.079)
		Climate diss. (temp.)	-0.235 (-0.295)	0.084 [Hnd; SC]
		Heating	-0.248	**0.029** [Hnd; SC]
		awMPD	-0.084 (-0.167)	0.236 [Hnd; SC]
		Maximum alpha PD diff.*Climate diss. (precip.)	0.702 (0.574)	**0.025** [Hnd; SC]
	Overwinter	Maximum alpha PD diff.	-0.191 (-0.117)	0.523 [Hnd; SC]
		Disturbed	0.099	0.690 [Hnd; SC]
		Climate diss. (precip.)	0.286 (0.237)	0.176 [Hnd; SC]
		Climate diss. (temp.)	-0.045 (-0.039)	0.825 [Hnd; SC]
		Heating	-0.152	0.222 [Hnd; SC]
		awMPD	0.105 (0.211)	0.196 [Hnd; SC]
	Second growing season	Maximum alpha PD diff.	0.021 (0.013)	0.958 [Hnd; SC]
		Disturbed	0.073	0.808 [Hnd; SC]
		Climate diss. (precip.)	0.458 (0.358)	0.155 [Hnd; SC]
		Climate diss. (temp.)	-0.063 (-0.059)	0.806 [Hnd; SC]
		Heating	-0.052	0.726 [Hnd; SC]
		awMPD	0.028 (0.056)	0.799 [Hnd; SC]
	Flowering*	Maximum alpha PD diff.	-0.123 (-0.097)	0.658 [Hnd; SC]
		Climate diss. (precip.)	0.324 (0.392)	**0.047** [Hnd; SC]
		Climate diss. (temp.)	-0.284 (-0.355)	0.118 [Hnd; SC]
		Heating	0.083	0.507 [Hnd; SC]

		awMPD	-0.114 (-0.229)	0.196 [Hnd; SC]

*Plants only flowered in disturbed plots, so these models exclude disturbance

Table S8: Summary of results, using *difference in median alpha phylogenetic diversity* (planted species PD minus abundance-weighted community PD) as response variable. Bold indicates significant p-value. In the p-value column, the superscript Hnd indicates the significance (either significant or not significant) was the same in a model using raw phylogenetic diversity (rather than the difference); the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Median alpha PD diff.	0.170 (0.116)	0.902 ^{SC}
		Disturbed	3.455 (3.668)	**<0.001** $^{Hnd; SC}$
		Climate diss. (precip.)	-1.162 (-1.628)	0.065 $^{Hnd; SC}$
		Climate diss. (temp.)	0.230 (0.343)	0.690 $^{Hnd; SC}$
		Heating	-0.861	**0.007** $^{Hnd; SC}$
		awMPD	-0.012 (-0.544)	0.103 $^{Hnd; SC}$
		Median alpha PD diff. * Disturbed	2.041 (1.385)	**0.039** $^{Hnd; SC}$
	First growing season	Median alpha PD diff.	-2.099 (-1.318)	0.317 $^{Hnd; SC}$
		Disturbed	-0.164 (0.581)	0.791 $^{Hnd; SC}$ (0.330)
		Climate diss. (precip.)	-0.763 (-0.890)	0.147 $^{Hnd; SC}$
		Climate diss. (temp.)	0.631 (0.747)	0.235 $^{Hnd; SC}$
		Heating	-1.003	**0.008** $^{Hnd; SC}$
		awMPD	0.005 (0.265)	0.589 $^{Hnd; SC}$
		Median alpha PD diff. * Disturbed	5.332 (3.350)	**0.011** $^{Hnd; SC}$
	Overwinter	Median alpha PD diff.	-3.712 (0.982)	**0.036** (0.205)
		Disturbed	1.362	**0.035** ^{Hnd}
		Climate diss. (precip.)	-1.728 (-1.933)	**0.020** ^{Hnd}
		Climate diss. (temp.)	-1.857 (-1.570)	**0.037**
		Heating	0.805	0.064 $^{Hnd; SC}$
		awMPD	-0.333 (-0.666)	0.240 $^{Hnd; SC}$
		Median alpha PD diff. * Climate diss. (temp.)	3.901 (3.035)	**0.050**
	Second growing season	Median alpha PD diff.	5.725 (3.272)	**0.034** $^{Hnd; SC}$
		Disturbed	0.594 (0.066)	0.417 $^{Hnd; SC}$ (0.932)
		Climate diss. (precip.)	-0.680 (-0.544)	0.360 ^{Hnd}
		Climate diss. (temp.)	0.089 (0.084)	0.864 $^{Hnd; SC}$

		Heating	-0.867	**0.046** Hnd; SC
		awMPD	-0.197 (-0.395)	0.428 Hnd; SC
		Median alpha PD diff. * Disturbed	-6.636 (-3.792)	**0.018** Hnd; SC
	Flowering*	Median alpha PD diff.	0.881 (0.601)	0.553 Hnd; SC
		Climate diss. (precip.)	-0.811 (-1.137)	0.351 Hnd; SC
		Climate diss. (temp.)	-0.483 (-0.719)	0.553 Hnd; SC
		Heating	-0.677	0.064 Hnd; SC
		awMPD	0.019 (0.038)	0.944 Hnd; SC
Number of plants in plot	Colonisation	Median alpha PD diff.	1.377 (0.835)	**0.004** Hnd; SC
		Disturbed	1.557	**<0.001** Hnd; SC
		Climate diss. (precip.)	-0.074 (-0.092)	0.737 Hnd
		Climate diss. (temp.)	-0.364 (-0.413)	0.121 Hnd; SC
		Heating	-0.318	**0.004** Hnd; SC
		awMPD	0.070 (0.141)	0.218 Hnd; SC
	First growing season	Median alpha PD diff.	0.498 (0.314)	0.090 Hnd; SC
		Disturbed	-0.050	0.772 Hnd; SC
		Climate diss. (precip.)	0.056 (0.065)	0.746 Hnd; SC
		Climate diss. (temp.)	-0.255 (-0.320)	0.079 Hnd; SC
		Heating	-0.241	**0.029** Hnd; SC
		awMPD	-0.090 (-0.179)	0.204 Hnd; SC
	Overwinter	Median alpha PD diff.	-0.022 (-0.013)	0.949 Hnd; SC
		Disturbed	0.123	0.617 Hnd; SC
		Climate diss. (precip.)	0.347 (0.287)	0.111 Hnd; SC
		Climate diss. (temp.)	-0.069 (-0.060)	0.732 Hnd; SC
		Heating	-0.149	0.238 Hnd; SC
		awMPD	0.119 (0.238)	0.147 Hnd; SC
	Second growing season	Median alpha PD diff.	-1.177 (0.085)	**0.009** SC (0.730)
		Disturbed	-0.072	0.812 Hnd; SC
		Climate diss. (precip.)	-1.028 (-0.585)	0.052 Hnd; SC
		Climate diss. (temp.)	0.196 (0.182)	0.318 Hnd; SC
		Heating	0.055	0.723 Hnd; SC
		awMPD	-0.050 (-0.101)	0.567 Hnd; SC
		Median alpha PD diff.*Climate diss. (precip.)	4.047 (1.799)	**<0.001** SC

	Flowering*	Median alpha PD diff.	0.093 (0.054)	0.816 [Hnd; SC]
		Climate diss. (precip.)	0.309 (0.374)	0.115 [Hnd; SC]
		Climate diss. (temp.)	-0.255 (-0.300)	0.177 [Hnd; SC]
		Heating	0.081	0.521 [Hnd; SC]
		awMPD	-0.095 (-0.190)	0.268 [Hnd; SC]

*Plants only flowered in disturbed plots, so these models exclude disturbance

Table S9: Summary of results, using *difference in gamma phylogenetic diversity* (planted species PD minus abundance-weighted community PD) as response variable. Bold indicates significant p-value. In the p-value column, the superscript Hnd indicates the significance (either significant or not significant) was the same in a model using raw phylogenetic diversity (rather than the difference); the superscript SC indicates the significance was the same in the models using single climate variables instead of principal components. In the effect size column, the effect size in brackets is the scaled effect size (centred divided by 2 standard deviations). Categorical variables were not scaled. Note that gamma PD was already scaled in the original analysis.

Response	Stage	Explanatory variable	Effect size	p-value
Any plants in plot	Colonisation	Gamma PD diff.	1.608	**<0.001** $^{Hnd; SC}$
		Disturbed	4.501	**<0.001** $^{Hnd; SC}$
		Climate diss. (precip.)	-0.608 (-0.851)	0.335 $^{Hnd; SC}$
		Climate diss. (temp.)	0.392 (0.583)	0.499 $^{Hnd; SC}$
		Heating	-0.838	**0.007** $^{Hnd; SC}$
		awMPD	-0.010 (-0.449)	0.170 $^{Hnd; SC}$
		Gamma PD diff. * Disturbed	-1.032	**<0.001** $^{Hnd; SC}$
	First growing season	Gamma PD diff.	0.515	0.105 $^{Hnd; SC}$
		Disturbed	0.578	0.322 $^{Hnd; SC}$
		Climate diss. (precip.)	-0.157 (-0.183)	0.762 $^{Hnd; SC}$
		Climate diss. (temp.)	0.699 (0.827)	0.190 $^{Hnd; SC}$
		Heating	-1.015	**0.007** $^{Hnd; SC}$
		awMPD	-0.040 (0.081)	0.867 $^{Hnd; SC}$
	Overwinter	Gamma PD diff.	0.492 (-0.299)	0.310 (0.459)
		Disturbed	1.378	**0.039**
		Climate diss. (precip.)	-3.286 (-3.673)	**0.001** $^{Hnd; SC}$
		Climate diss. (temp.)	-0.792 (-0.962)	0.192 $^{Hnd; SC}$
		Heating	0.807	0.063 ^{SC}
		awMPD	-0.397 (-0.794)	0.162 $^{Hnd; SC}$
		Gamma PD diff. * Climate diss. (precip.)	-1.617 (-1.808)	**0.025** ^{Hnd}
	Second growing season	Gamma PD diff.	0.131	0.643 $^{Hnd; SC}$
		Disturbed	0.817	0.232 $^{Hnd; SC}$
		Climate diss. (precip.)	-0.596 (-0.478)	0.375 $^{Hnd, SC}$
		Climate diss. (temp.)	0.139 (0.133)	0.769 $^{Hnd; SC}$
		Heating	-0.753	0.069 $^{Hnd; SC}$
		awMPD	-0.061 (-0.122)	0.782 $^{Hnd; SC}$
	Flowering*	Gamma PD diff.	0.404	0.262 $^{Hnd; SC}$

		Climate diss. (precip.)	-0.551 (-0.772)	0.519 [Hnd; SC]
		Climate diss. (temp.)	-0.421 (-0.626)	0.704 [Hnd; SC]
		Heating	-0.615	0.079 [Hnd; SC]
		awMPD	0.056 (0.113)	0.835 [Hnd; SC]
Number of plants in plot	Colonisation	Gamma PD diff.	0.024	0.840 [Hnd; SC]
		Disturbed	1.562	**<0.001** [Hnd; SC]
		Climate diss. (precip.)	0.115 (0.144)	0.616 [Hnd; SC]
		Climate diss. (temp.)	-0.422 (-0.479)	0.111 [Hnd; SC]
		Heating	-0.348	**0.003** [Hnd; SC]
		awMPD	0.050 (0.099)	0.392 [Hnd; SC]
	First growing season	Gamma PD diff.	0.146	0.126 [Hnd; SC]
		Disturbed	0.023	0.897 [Hnd; SC]
		Climate diss. (precip.)	0.210 (0.243)	0.176 [Hnd; SC]
		Climate diss. (temp.)	-0.277 (-0.285)	0.118 [Hnd; SC]
		Heating	-0.240	**0.036** [Hnd; SC]
		awMPD	-0.084 (-0.167)	0.240 [Hnd; SC]
	Overwinter	Gamma PD diff.	0.032	0.747 [Hnd; SC]
		Disturbed	0.142	0.573 [Hnd; SC]
		Climate diss. (precip.)	0.369 (0.305)	0.088 [Hnd; SC]
		Climate diss. (temp.)	-0.074 (-0.065)	0.710 [Hnd; SC]
		Heating	-0.144	0.250 [Hnd; SC]
		awMPD	0.122 (0.245)	0.123 [Hnd; SC]
	Second growing season	Gamma PD diff.	0.175	0.141 [Hnd; SC]
		Disturbed	0.159	0.588 [Hnd; SC]
		Climate diss. (precip.)	0.612 (0.478)	**0.037** [Hnd; SC]
		Climate diss. (temp.)	-0.096 (-0.089)	0.678 [SC]
		Heating	-0.049	0.742 [Hnd; SC]
		awMPD	0.050 (0.100)	0.621 [Hnd; SC]
	Flowering*	Gamma PD diff.	-0.005	0.960 [Hnd; SC]
		Climate diss. (precip.)	0.332 (0.401)	**0.048** [Hnd; SC]
		Climate diss. (temp.)	-0.269 (-0.317)	0.155 [Hnd; SC]
		Heating	0.078	0.533 [Hnd; SC]
		awMPD	-0.100 (-0.200)	0.233 [Hnd; SC]

*Plants only flowered in disturbed plots, so these models exclude disturbance

Supplementary tables for species origin and traits

Table S10: Difference in model performance when adding species status (native/non-native) and biogeographic origin (native continent syndrome of Fristoe et al. 2023) into the main models testing the effect of PD. Positive values of ΔAIC indicate the model is *worse* when including species status or origin.

Research question	Model	ΔAIC when adding native/exotic	ΔAIC when adding origin	Affect significance or direction of PD?
Question 1 (absolute PD)	Colonisation (any plants in plot)	-0.4	2.0	No
	First growing season (any plants in plot)	2.0	1.9	No
	Overwinter (any plants in plot)	-3.2	2.0	No
	Second growing season (any plants in plot)	-2.3	1.2	No
	Colonisation (number of plants)	-2.2	1.9	No
	First growing season (number of plants)	1.7	-0.9	Yes*
	Overwinter (number of plants)	1.8	2.0	No
	Second growing season (number of plants)	0.3	-1.5	No
Question 2 (relative PD)	Colonisation (any plants in plot)	1.2	2.0	No
	First growing season (any plants in plot)	1.3	2.0	No
	Overwinter (any plants in plot)	0.7	-0.4	No
	Second growing season (any plants in plot)	0.7	2.0	No
	Colonisation (number of plants)	-5.1	2.0	No
	First growing season (number of plants)	1.5	-0.3	No
	Overwinter (number of plants)	1.7	2.0	No
	Second growing season (number of plants)	1.4	0.1	No

*Including biogeographic origin removed the PD*disturbance interaction, but the main effect of PD was still positive and significant ($p = 0.043$).

Table S11: Difference in structural equation model performance when adding the traits [SLA + height] or [SLA + height + seed mass]. AIC values were compared between models including only the indirect effects of indigenous-PD (i.e. the effect of indigenous-PD is mediated entirely through its effect on the included traits; Fig S8b), versus models that include both indirect effects and direct effects of indigenous-PD (Fig. S8a). Positive values of ΔAIC indicate the model is *worse* when including indirect effects of indigenous-PD only; in other words, positive values of ΔAIC indicate Fig. S8a is preferred. Only models where indigenous-PD was found to be significant in the main models were tested (see Fig. 3b-d and Fig. 5a-d in main text); note that models from the second year where indigenous-PD was found to be significant are not included due to too few data to appropriately parameterise SEMs.

Research question	**Model**	**Traits included**	**ΔAIC when only including indirect effects (effect of PD entirely through traits)**	**PD p-value when including traits**
Question 1 (absolute PD)	First growing season (any plants in plot)	SLA + Height	6.2	**0.028**
		SLA + Height + Seed mass	13.0	**0.001**
	Colonisation (number of plants)	SLA + Height	-1.5	**<0.001**
		SLA + Height + Seed mass	1.3	**<0.001**
	First growing season (number of plants)	SLA + Height	6.2	**<0.001**
		SLA + Height + Seed mass	7.4	**<0.001**
Question 2 (relative PD)	Colonisation (any plants in plot)	SLA + Height	2.5	0.103*
		SLA + Height + Seed mass	2.4	0.111*
	First growing season (any plants in plot)	SLA + Height	5.4	**0.025**
		SLA + Height + Seed mass	5.1	**0.029**
	Colonisation (number of plants)	SLA + Height	2.9	0.087*
		SLA + Height + Seed mass	2.2	0.124*
	First growing season (number of plants)	SLA + Height	1.3	0.189*
		SLA + Height + Seed mass	1.5	0.172*

*While PD was no longer significant in these models, this was a product of the lower number of datapoints necessitated by only including species with relevant trait data – these values were also non-significant when model structure was the same as the main models i.e. not including traits at all.

Table S12: Relationship between traits and indigenous-PD for all four PD metrics. Competition index and resistance to herbivory was only available for 62 and 63 of the 93 species included in the Kempel study. Superscript numbers refer to data sources which follow at the end of the table. Three tests per PD metric were performed: using all available data ("all data"); limiting data to species that were native to Europe or North America ("EU/NA"); and including biogeographic syndrome as a random variable ("BG RV").

Trait	Number of species	Metric	p-value (all data)	p-value (EU/NA)	p-value (BG RV)
Seed mass (mg)	153 [1, 2, 3]	Mean alpha PD	**0.036**	0.053	**0.028**
		Maximum alpha PD	**0.042**	**0.049**	**0.039**
		Median alpha PD	**0.024**	**0.009**	**0.022**
		Gamma PD	0.694	0.998	0.754
Optimal germination rate	151 [1, 2]	Mean alpha PD	0.180	0.192	0.180
		Maximum alpha PD	0.749	0.619	0.749
		Median alpha PD	0.955	0.932	0.955
		Gamma PD	0.172	0.097	0.172
Competition index	62 [2]	Mean alpha PD	0.516	0.841	0.507
		Maximum alpha PD	0.491	0.681	0.478
		Median alpha PD	0.351	0.694	0.337
		Gamma PD	0.435	0.524	0.424
Resistance to herbivory	63 [2]	Mean alpha PD	0.107	0.111	0.111
		Maximum alpha PD	0.363	0.499	0.416
		Median alpha PD	0.780	0.567	0.760
		Gamma PD	0.571	0.685	0.479
Hardiness Index	63 [1]	Mean alpha PD	0.396	0.271	0.674
		Maximum alpha PD	0.186	**0.037**	0.129
		Median alpha PD	**0.014**	0.109	**0.029**
		Gamma PD	0.190	0.158	**0.042**
Life history (annual/perennial)	63 [1]	Mean alpha PD	0.803	0.780	0.637
		Maximum alpha PD	0.267	0.953	0.246
		Median alpha PD	0.710	0.353	0.848
		Gamma PD	0.212	0.252	0.216
Vegetative height (m)	149 [3, 4]	Mean alpha PD	0.185	0.279	0.166
		Maximum alpha PD	**0.001**	**<0.001**	**0.001**
		Median alpha PD	0.259	0.819	0.126
		Gamma PD	0.137	0.073	0.184

Generative height (m)	103 [3]	Mean alpha PD	0.533	0.594	0.514
		Maximum alpha PD	**0.001**	**<0.001**	**0.005**
		Median alpha PD	0.421	0.251	0.922
		Gamma PD	0.280	0.144	0.619
Leaf nitrogen (mg/g)	70 [3]	Mean alpha PD	**0.038**	**0.010**	**0.032**
		Maximum alpha PD	0.788	0.596	0.751
		Median alpha PD	0.163	0.270	0.353
		Gamma PD	**0.030**	**0.014**	0.111
SLA (mm^2/mg)	133 [3, 5, 6, 7, 8]	Mean alpha PD	0.134	0.257	0.129
		Maximum alpha PD	0.190	0.243	0.184
		Median alpha PD	0.835	0.750	0.833
		Gamma PD	**0.006**	**0.025**	**0.005**
LDMC (g/g)	100 [3]	Mean alpha PD	**0.012**	**0.013**	**0.011**
		Maximum alpha PD	0.686	0.754	0.899
		Median alpha PD	**0.023**	0.062	**0.019**
		Gamma PD	0.504	0.593	0.530

[1] Haeuser et al. 2017
[2] Kempel et al. 2013
[3] Kattge et al. 2020
[4] Falster et al. 2021
[5] Wang et al. 2017
[6] Anten & Hirose 1999
[7] Elkins & van Iersel 2020
[8] Wang et al. 2024

Supplementary References

Anten, N. P., & Hirose, T. (1999). Interspecific differences in above-ground growth patterns result in spatial and temporal partitioning of light among species in a tall-grass meadow. *Journal of Ecology*, *87*(4), 583-597.

Belsky, A. J. (1994). Influences of trees on savanna productivity: tests of shade, nutrients, and tree-grass competition. *Ecology*, *75*(4), 922-932.

BGCI (2026). GlobalTreeSearch. Botanic Gardens Conservation International. Richmond, U.K. Available at https://tools.bgci.org/global_tree_search.php. Accessed on 16/02/2016.

Bond, W. J., & Midgley, G. F. (2012). Carbon dioxide and the uneasy interactions of trees and savannah grasses. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *367*(1588), 601-612.

Carmona, C. P., Bueno, C. G., Toussaint, A., Träger, S., Díaz, S., Moora, M., ... & Tamme, R. (2021). Fine-root traits in the global spectrum of plant form and function. *Nature*, *597*(7878), 683-687.

Czyżewski, S., Søndergaard, S. A., Molnár, Á. P., Kerr, M. R., Kristensen, J. A., Atkinson, J., ... & Svenning, J. C. (2026). Revisiting Europe's temperate forests: Palaeoecological evidence for an herbivory-driven woodland-grassland mosaic biome. *Biological Conservation*, *316*, 111749.

Díaz, S., Kattge, J., Cornelissen, J. H., Wright, I. J., Lavorel, S., Dray, S., ... & Gorné, L. D. (2016). The global spectrum of plant form and function. *Nature*, *529*(7585), 167-171.

Dohn, J., Dembélé, F., Karembé, M., Moustakas, A., Amévor, K. A., & Hanan, N. P. (2013). Tree effects on grass growth in savannas: competition, facilitation and the stress-gradient hypothesis. *Journal of Ecology*, *101*(1), 202-209.

Dullinger, I., Wessely, J., Bossdorf, O., Dawson, W., Essl, F., Gattringer, A., ... & Dullinger, S. (2017). Climate change will increase the naturalization risk from garden plants in Europe. *Global Ecology and Biogeography*, *26*(1), 43-53.

Elkins, C., & van Iersel, M. W. (2020). Longer photoperiods with the same daily light integral improve growth of rudbeckia seedlings in a greenhouse. *HortScience*, *55*(10), 1676-1682.

Falster, D., Gallagher, R., Wenk, E. H., Wright, I. J., Indiarto, D., Andrew, S. C., ... & O'sullivan, O. S. (2021). AusTraits, a curated plant trait database for the Australian flora. *Scientific Data*, *8*(1), 254.

Fristoe, T. S., Bleilevens, J., Kinlock, N. L., Yang, Q., Zhang, Z., Dawson, W., ... & van Kleunen, M. (2023). Evolutionary imbalance, climate and human history jointly shape the global biogeography of alien plants. *Nature Ecology & Evolution*, *7*(10), 1633-1644.

Gelman, A. (2008). Scaling regression inputs by dividing by two standard deviations. *Statistics in Medicine*, *27*(15), 2865-2873.

Gorné, L. D., Aguirre-Gutiérrez, J., Souza, F. C., Swenson, N. G., Kraft, N. J. B., Schwantes Marimon, B., ... & Díaz, S. (2025). Use and misuse of trait imputation in ecology: the problem of using out-of-context imputed values. *Ecography*, *2025*(6), e07520.

Grenié, M., Bruelheide, H., Dawson, W., Essl, F., van Kleunen, M., Kühn, I., ... & Winter, M. (2025). Gaps in global alien plant trait data and how to fill them. *Global Ecology and Biogeography*, *34*(10), e70131.

Haeuser, E., Dawson, W., & van Kleunen, M. (2017). The effects of climate warming and disturbance on the colonization potential of ornamental alien plant species. *Journal of Ecology*, *105*(6), 1698-1708.

Kattge, J., Bönisch, G., Díaz, S., Lavorel, S., Prentice, I. C., Leadley, P., ... & Cuntz, M. (2020). TRY plant trait database–enhanced coverage and open access. *Global Change Biology, 26*(1), 119-188.

Kempel, A., Chrobock, T., Fischer, M., Rohr, R. P., & van Kleunen, M. (2013). Determinants of plant establishment success in a multispecies introduction experiment with native and alien species. *Proceedings of the National Academy of Sciences*, 110(31), 12727-12732.

Lefcheck, J. S. (2016). piecewiseSEM: Piecewise structural equation modelling in r for ecology, evolution, and systematics. *Methods in Ecology and Evolution*, *7*(5), 573-579.

Ludwig, F., Dawson, T. E., Prins, H. H. T., Berendse, F., & De Kroon, H. (2004). Below-ground competition between trees and grasses may overwhelm the facilitative effects of hydraulic lift. *Ecology letters*, *7*(8), 623-631.

Moles, A. T., Ackerly, D. D., Tweddle, J. C., Dickie, J. B., Smith, R., Leishman, M. R., ... & Westoby, M. (2007). Global patterns in seed size. *Global Ecology and Biogeography*, *16*(1), 109-116.

Revell, L. J. (2024). phytools 2.0: an updated R ecosystem for phylogenetic comparative methods (and other things). *PeerJ*, *12*, e16505.

Riginos, C. (2009). Grass competition suppresses savanna tree growth across multiple demographic stages. *Ecology*, *90*(2), 335-340.

Scholes, R. J., & Archer, S. R. (1997). Tree-grass interactions in savannas. *Annual Review of Ecology and Systematics*, *28*(1), 517-544.

Silander Jr, J. A., Bond, W. J., & Ratsirarson, J. (2024). The grassy ecosystems of Madagascar in context: Ecology, evolution, and conservation. *Plants, People, Planet*, *6*(1), 94-115.

Smith, S. A., & Brown, J. W. (2018). Constructing a Broadly Inclusive Seed Plant Phylogeny. *American Journal of Botany*, *105*, 302–314.

Stachowicz, J. J., & Tilman, D. (2005). Species invasions and the relationships between species diversity, community saturation, and ecosystem functioning. *Species invasions: insights into ecology, evolution, and biogeography*, 41-64.

Strömberg, C. A. (2011). Evolution of grasses and grassland ecosystems. *Annual Review of Earth and Planetary Sciences*, *39*(1), 517-544.

van Kleunen, M., Xu, X., Yang, Q., Maurel, N., Zhang, Z., Dawson, W., ... & Fristoe, T. S. (2020). Economic use of plants is key to their naturalization success. *Nature Communications*, *11*(1), 3201.

Wang, C., Zhou, J., Liu, J., & Jiang, K. (2017). Differences in functional traits between invasive and native *Amaranthus* species under different forms of N deposition. *The Science of Nature*, *104*(7), 59.

Wang, F., Zhang, Q., Huang, P., Li, C., & Li, Y. (2024). CSR strategy composition and leaf traits for herbaceous plants in garden design. *Ecological Indicators*, *158*, 111173.
Yang, Q., Weigelt, P., Fristoe, T. S., Zhang, Z., Kreft, H., Stein, A., ... & van Kleunen, M. (2021). The global loss of floristic uniqueness. *Nature Communications*, *12*(1), 7290.